

HGMAP

Digital Protection Relay

HD HYUNDAI ELECTRIC

Essential for Today, Potential for Tomorrow

HD Hyundai Electric solely pursues the growth of our customers' business. From power generation to power distribution, we focus on developing and commercializing products and solutions aimed at increasing the efficiency of energy equipment as well as at proactively monitoring and controlling assets in an integrated manner to improve our customers' productivity and management efficiency. We are well aware that our efforts add to the driving force behind our customers' growth and contribute to the creation and maintenance of a more dynamic world. We focus on achieving innovation and strive to evolve continuously to shape a better tomorrow based on today's technological advancements.

Selection Table

Digital Protection Relay (HGMAP-BCS)

Type		Specification
General		
Application		Provide optimized protection and monitoring for feeder/Incoming and medium-voltage motors in plant, factory below 25kV and measures electrical quantities, and controls digital input/output contacts.Communication with supervision systems such as SCADA enables remote device control and provides real-time status and measurements.
Environment	Temperature	Rated ambient temperature : 25°C Continuous operation temperature : -10 to 55°C Storage temperature : -20 to 75°C
	Humidity	Under 80%, Condensation must be avoided
	Location	Location below 2,000m altitude Not exposed to direct sunlight, snow, rain, wind, etc Not exposed to corrosive, explosive gas and dust
Configuration		
Input	Control power	AC/DC 88 – 265V
	Analog input	PT 4 channel, CT 8 channel
Protection		OCR1-I(50-1), OCR1-INV(51-1), OCR2-I(50-1), OCR2-INV(51-1), OCGR-I(50N), OCGR-INV(51N), SGR(67G), NSOCR(46), THR(49), UCR(37), STALL/LOCK(48/51LR), NCH(66), DOCR-I(67-I), DOCR-INV(67-D), DOCGR-I(67N-I), DOCGR-INV(67N-D), DFR(87), OVR(59), UVR(27), UVR-X(27R), OVGR(64), POR(47P), NSOVR(47N), SYNC_CHK(25), RCLS(79), APR(32P), RPR(32Q), UPR(37P), OFR(81O), UFR(81U), FROC(81R), OEX(24), LOF(40Q), OST(78S), INRS(95I)
Measure		Voltage (Line, Phase, Positive, Negative and Zero Sequence), Current (Phase, Positive, Negative and Zero Sequence), Power, Energy (Active, Reactive) Phase, Frequency, Power Factor, Harmonics Demand, Load rate
Display	Screen	7.0inch Color TFT LCD (capacitive touchscreen)
	LED	Status & Alarm LEDs×23
Event log	Event	max. 2048 records
	PQ event	max. 256 records
	Fault event	max. 128 records
	Fault wave	max. 200 records (IEEE C37.111 Comtrade format)
Size	Dimension	227×278×207.5mm (W×H×D)
	Weight	6.8kg (Gross weight)

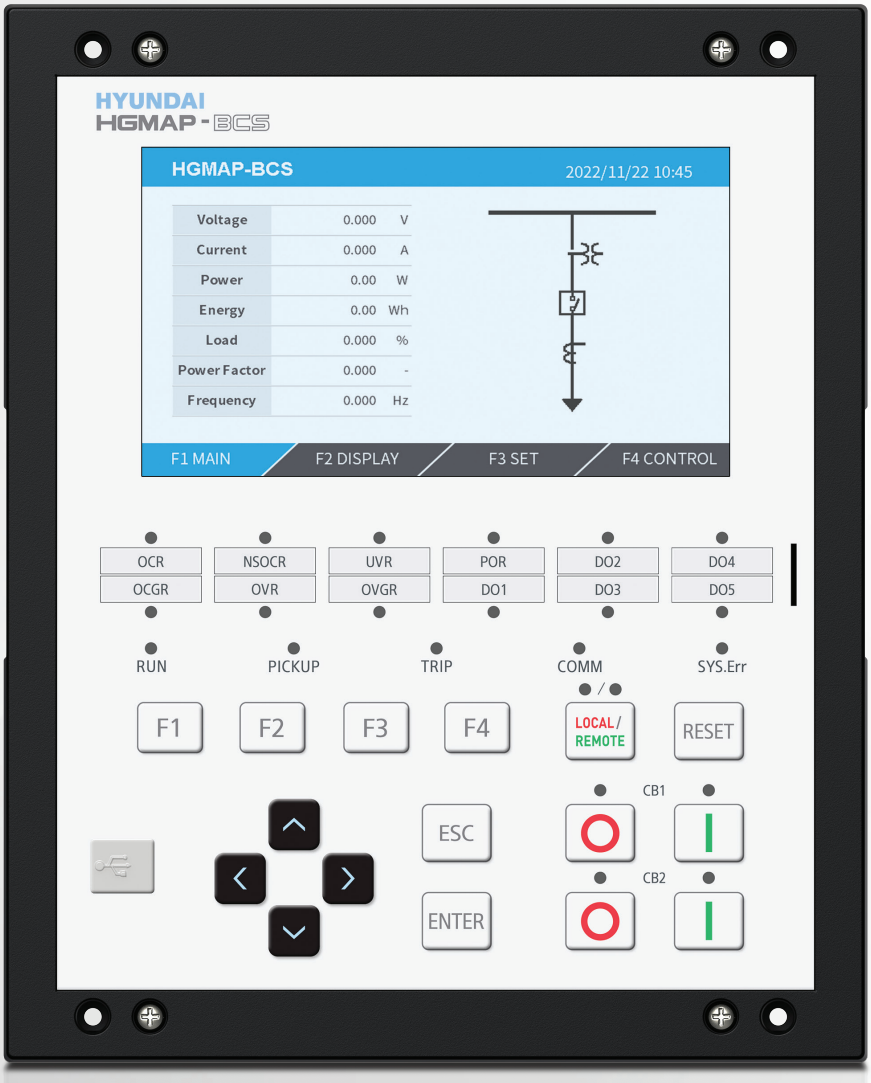
Type			Specification
Rating			
Control power	Rated power		AC/DC 110/220V
	Input range		AC/DC 88 – 265V
	Voltage sag immunity		100ms (at 110Vdc)
	Consumption		Standby 15W or less, Operation 30W or less
CT input	CT primary set		5 – 9,000A or 1 – 9,000A
	CT secondary set		5A or 1A (1In)
	Maximum load		below 1.0VA at 1In
	Input range		0 – 40In
	Overload capability		2In(continuous), 20In(for 2s), 40In(for 50ms)
PT input	PT primary set		110 – 345,000V
	PT secondary set		110/√3 –190V (Vn)
	PT tertiary set		100 – 190V (Von)
	Maximum load		below 0.5VA at 1Vn
	Input range		0 – 2.0Vn
	Overload capability		1.2Vn(continuous), 2Vn(for 1hour)
Digital Input (DI)	Rated input		110Vdc
	Threshold voltage		Turn-on 80VDC, Turn-off 70VDC
	Input delay		within 10ms
	Maximum load		0.5VA at 1Vn
Digital Output (DO) for control	Contact type		Dry contact
	Contact rating		Resistive load : 10A at 250VAC/30VDC
			Inductive load : 5A at 250VAC/30VDC
	Operating speed		within 10ms
Digital Output (DO) for signal	Contact type		Dry contact
	Contact rating		Resistive load : 5A at 250VAC/30VDC
			Inductive load : 2A at 250VAC/30VDC
	Operating speed		within 10ms
Communication	RS-485	Wiring	2-wire(D+,D-), 4-wire(Rx+,Rx-,Tx+,Tx-)
		Baud rate	960, 19200, 38400, 57600 bps
		Protocol	Modbus/RTU, DNP3.0
	Ethernet	Port	LAN(RJ45), Fiber optic(SFP multi-mode LC type), HSR/PRP
		Speed	10/100Mbps
		Protocol	Modbus/TCP, IEC 61850 Ed.2
	Manager	Connector	USB mini-B port
	Software	Protocol	Reserved
Type Test			
Certificate	Domestic		KEMC 1120-0579:2018
EMC	RF immunity		IEC 61000-4-3 / IEC 60255-26
	Conducted RF		IEC 61000-4-6 / IEC 60255-26
	ESD		IEC 61000-4-2 / IEC 60255-26
	Fast transient		IEC 61000-4-4 / IEC 60255-26
	Surge immunity		IEC 61000-4-5 / IEC 60255-26
	Power magnetic		IEC 61000-4-8 / IEC 60255-26
	Voltage dips, interruptions		IEC 61000-4-11 / IEC 60255-26, IEC 61000-4-29 / IEC 60255-26
	RF emission		CISPR 11,22 / IEC 60255-26
Mechanical	Vibration		IEC 60255-21-1 / IEC 60255-27
	Shock/Bump		IEC 60255-21-2 / IEC 60255-27
	Seismic		IEC 60255-21-3 / IEC 60255-27
Environment	Dry heat		IEC 60068-2-2 / IEC 60255-27
	Cold		IEC 60068-2-1 / IEC 60255-27
	Damp-heat		IEC 60068-2-78 / IEC 60255-27
	Cyclic temperature		IEC 60068-2-30 / IEC 60255-27
Withstand	Insulation Resistance		IEC 60255-5 / IEC 60255-27
	Dielectric		IEC 60255-5 / IEC 60255-27
	Impulse voltage		IEC 60255-5 / IEC 60255-27

HG Series

Digital Protection Relay

HGMAP-BCS

Best Solution for Stability for Power Distribution System
An Exceptional Power Monitoring and Control Solution Delivering Both Economy and Convenience



Intelligent digital multifunctional protection relay, HGMAP-BCS, is designed to meet the growing demand in the 22.9 kV distribution switchgear market and the advancing requirements of transmission and distribution automation. Equipped with expandable I/O and IEC 61850 communication capabilities, it is optimized for system monitoring, protection, and sequence control in plant and maritime environments.

Enhanced Protection and Measurement Performance

- Integrated protection for Feeder, Motor, TR, DG, and Generator
- Measurement of current and power demand, including up to the 15th harmonic
- Power quality monitoring functions (Sag, Swell, Interruption)

Improved User Interface

- 7-inch TFT-LCD color display
- Intuitive touch-based operation
- Compatible with the device management software (HG Manager)

I/O Expandability

- Configurable I/O options tailored to customer requirements
- Expansion options for additional DI/DO, AI, and RTD

IEC 61850 Compliance

- Compliant with the international IEC 61850 standard for power system communication
- Equipped with IEC 61850 Edition 2.0 communication stack
- Supports high-speed redundant communication (HSR/PRP)



Overview

Digital Protection Relay (HGMAP-BCS)

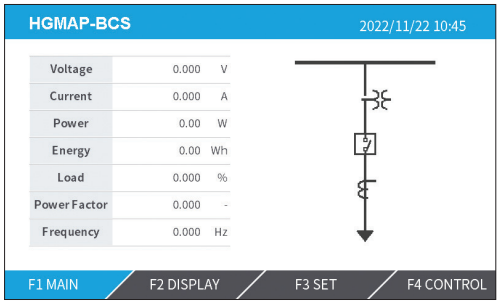


Enhanced Protection and Measurement Performance

- Provides robust and integrated protection for a wide range of load environments, including Feeder/Incoming lines, medium- and low-voltage motors, two-winding transformers, and renewable distributed energy resources such as photovoltaic generation systems and maritime.
- Supports real-time monitoring of harmonics up to the 15th order, as well as current and power demand measurement.
- Records up to 2,048 operational events and up to 128 fault events, enabling efficient root-cause analysis and prompt corrective action in the event of a system disturbance.

User-Friendly Interface

- Equipped with a 7-inch TFT-LCD color display with 800 × 480 resolution, providing clear, detailed, and accurate data visibility at a glance.
- Offers an intuitive capacitive touch interface with fast response and reduced operator fatigue during operation. When touch input is impractical, physical buttons are also available to ensure stable and reliable device operation.



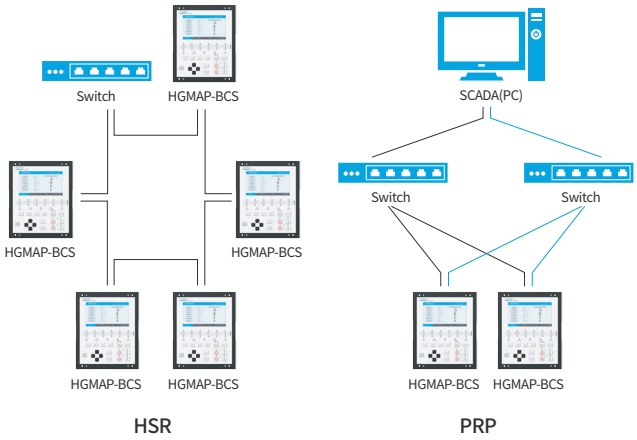
IEC 61850 Communication Compliance

- IEC 61850 is an international standard communication protocol widely adopted across transmission, distribution, and substation automation systems.
- HGMAP-BCS incorporates the IEC 61850 Edition 2.0 communication stack, enabling it to operate as an IED within automated substations and to perform real-time system monitoring and control via MMS and GOOSE communication with SCADA systems or other IEDs.



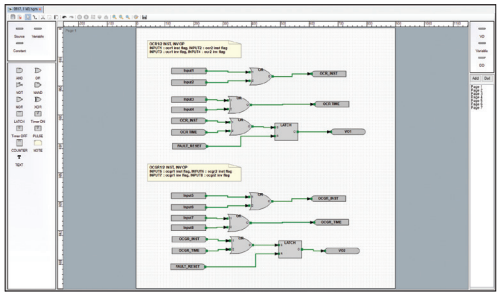
High-Speed Redundant Communication Support

- Supports high-speed redundant communication based on HSR (High-Availability Seamless Redundancy) and PRP (Parallel Redundancy Protocol), ensuring enhanced reliability and stability for automation networks.



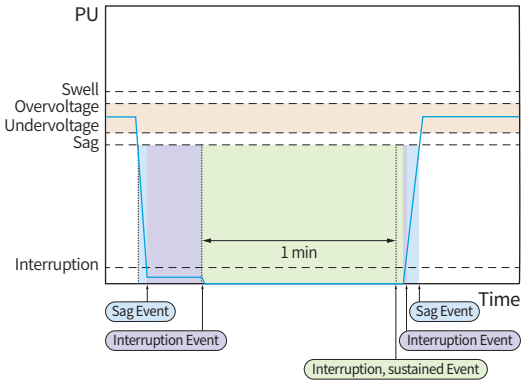
User Logic and Single-Line Diagram Functions

- Allows users to freely configure and apply custom logic through a dedicated PLC logic editing software.
- The manager software enables easy creation and modification of the system single-line diagram, which is continuously displayed on the main interface.



PQ Monitoring and RMS Trend Functions

- Provides power quality (PQ) monitoring features for sag, swell, and interruption events, and records up to 256 PQ events separately to support stable power quality management.
- Records RMS trends in 15-minute intervals for system analysis and operational insights.

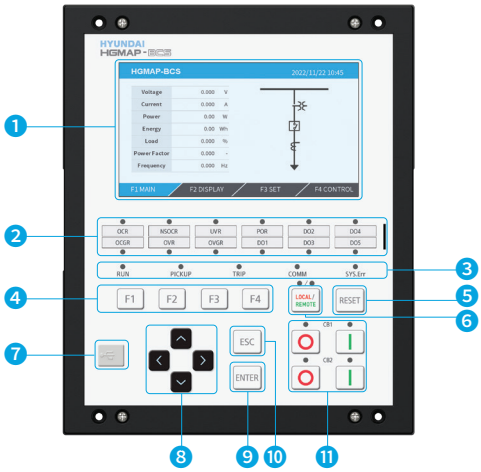


Appearance

Digital Protection Relay (HGMAP-BCS)

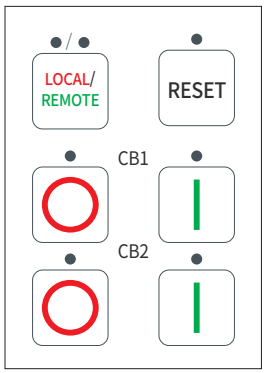
Front View

No.	Classification	Description
1	Graphic LCD	Display measurement, setting and record with touch control
2	TRIP / DO Alarm LED	User-defined TRIP and DO state
3	Status LED	Display the status of device and pick-up state
4	Hotkey	F1–F4 menu hotkey
5	RESET button	Cancel the TRIP alarm, LED and DO reset control
6	Local/Remote button	Local/remote control to toggle permission
7	USB connector port	mini-B USB port for connection for maintenance
8	Arrow keys	Move between pages, change setting value
9	ENTER button	Enter menu page, confirm save
10	ESC button	Return to previous page, cancel save
11	Breaker control button	Control the breaker open/close



LED Status Guide

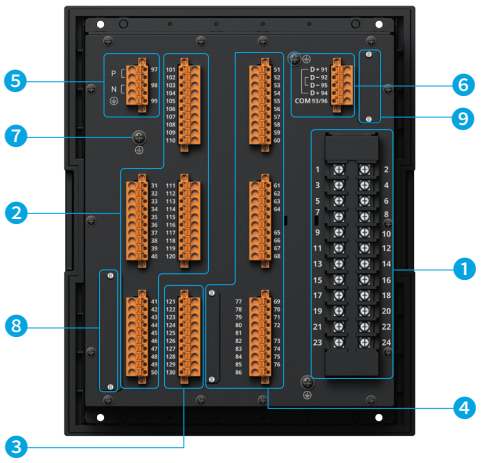
Classification		LED Status	Description	Note
Status LED	RUN	●	Normal operation	
	PICKUP	◐ Blinking	Pick-up for protection relay	
		●	The TRIP operates after Pick-up	
	TRIP	●	Relay TRIP operates	
	COMM	◐ Blinking	Communication	
	SYS.Err	●	System failure	
TRIP alarm LED	#1–#12	●	Each TRIP operates	User-defined



Classification		LED Status	Description
Status LED	O	●	Breaker OPEN
	I	●	Breaker CLOSE
	LOCAL	●	LOCAL control established
	REMOTE	●	REMOTE control established

T/B Rear View

No.	Classification	Terminal No.	Description
1	PT/CT input terminal	X1:1–10	CT pri. terminal(Ia, Ib, Ic, In, Io)
		X1:11–16	CT sec. terminal(Ia, Ib, Ic)
		X1:17–24	PT terminal (Va, Vb, Vc)
2	DI input terminal	X7:31–40	DI01–DI08
		X7:41–50	DI09–DI16
3	DO output terminal	X3:51–60	DO01–DO06
		X4:61–64	DO07–DO08
		X4:65–68	TCS1, TCS2
		X5:69–76	CB1 ON/OFF, CB2 ON/OFF
4	Optional I/O	X9:101–110	DI17–DI24
		X10:111–117	DI25–DI28
		X11:121–130	DO13–DO22
		X12:77–86	[DIO] DI29–DI32, DO23–DO26
			[AI] AI1–AI4
			[RTD] RTD1–RTD3
5	Control power terminal	X6:97–99	Control power P, N, FG
6	Modbus comm. terminal	X2:91–96	X2:91–96 : D+, D-, D-, D+, COM
7	Ground terminal	⏚	Ground connection
8	IEC 61850 comm.	X13	IEC 61850 communication
9	DNP3.0 comm.	X14	DNP3.0 communication (reserved)



Contact Configuration

X6	
97	P
97	P
98	N
98	N
99	F.G

X9	
101	DI 17
102	DI 18
103	DI 19
104	DI 20
105	DI COM(17–20)
106	DI 21
107	DI 22
108	DI 23
109	DI 24
110	DI COM(21–24)

X12 (RTD option)	
77	RTD1-A
78	RTD1-B
79	RTD1-C
80	RTD2-A
81	RTD2-B
82	RTD2-C
83	RTD3-A
84	RTD3-B
85	RTD3-C
86	F.G

X3	
51	DO 01
52	DO 02
53	DO 03
54	DO COM(01–03)
55	DO 04
56	DO COM(04)
57	DO 05
58	DO COM(05)
59	DO 06
60	DO COM(06)

X2, X14	
91	RXP(D+)
92	RXN(D-)
95	TXN(D-)
94	TXP(D+)
93/96	COM

X7	
31	DI 01
32	DI 02
33	DI 03
34	DI 04
35	DI COM(01–16)
36	DI 05
37	DI 06
38	DI 07
39	DI 08
40	DI COM(01–16)

X10	
111	DI 25
112	DI 26
113	DI 27
114	DI 28
115	DI COM(25–28)
116	DI COM(25–28)
117	DI COM(25–28)
118	N.C
119	DO COM(13–22)
120	DO COM(13–22)

X12 (AI option)	
77	AI 01
78	AI COM(01)
79	AI 02
80	AI COM(02)
81	AI 03
82	AI COM(03)
83	AI 04
84	AI COM(04)
85	N.C
86	F.G

X4	
61	DO 07
62	DO COM(07)
63	DO 08
64	DO COM(08)
65	TCS1+
66	TCS1-
67	TCS2+
68	TCS2-

X1			
1	Ia_Pri	2	Ia_Pri
3	Ib_Pri	4	Ib_Pri
5	Ic_Pri	6	Ic_Pri
7	I_N	8	I_N
9	Io	10	Io
11	Ia_Sec	12	Ia_Sec
13	Ib_Sec	14	Ib_Sec
15	Ic_Sec	16	Ic_Sec
17	Va	18	Va
19	Vb	20	Vb
21	Vc	22	Vc
23	Vo	24	Vo

X8	
41	DI 09
42	DI 10
43	DI 11
44	DI 12
45	DI COM(01–16)
46	DI 13(CB2 52a)
47	DI 14(CB2 52b)
48	DI 15(CB1 52a)
49	DI 16(CB1 52b)
50	DI COM(01–16)

X11	
121	DO 13
122	DO 14
123	DO 15
124	DO 16
125	DO 17
126	DO 18
127	DO 19
128	DO 20
129	DO 21
130	DO 22

X12 (DIO option)	
77	DI 29
78	DI 30
79	DI 31
80	DI 32
81	DI COM(29–32)
82	DO 23
83	DO 24
84	DO 25
85	DO 26
86	DO COM(23–26)

X5	
69	CB2 CLOSE
70	DO COM(09)
71	CB2 OPEN
72	DO COM(10)
73	CB1 CLOSE
74	DO COM(11)
75	CB1 OPEN
76	DO COM(12)

Technical Data

Measurement

Item		Range [Unit]	Accuracy	Note
Voltage	Phase voltage	0.000 – 999.99 [kV]	±0.2%	Va(PT1), Vb(PT2), Vc(PT3)
	Zero phase voltage	0.000 – 999.99 [kV]	±0.2%	Vo(PT4)
	Line voltage 1)	0.000 – 999.99 [kV]	±0.2%	Vab, Vbc, Vca
	Positive sequence	0.000 – 999.99 [kV]	±1.0%	V1
	Negative sequence	0.000 – 999.99 [kV]	±1.0%	V2
	Zero sequence	0.000 – 999.99 [kV]	±1.0%	V0
	Unbalance rate	0.00 – 100.0 [%]	±1.0%	Vu
Current	Phase current	0.000 – 999.99 [kA]	±0.2%	Ia(CT1), Ib(CT2), Ic(CT3)
	Zero phase current	0.000 – 999.99 [kA]	±0.2%	Io(CT4)
	Positive sequence	0.000 – 999.99 [kA]	±1.0%	I1
	Negative sequence	0.000 – 999.99 [kA]	±1.0%	I2
	Zero sequence	0.000 – 999.99 [kA]	±1.0%	I0
Phase	Voltage	0.0 – 360.0 [°]	±1.0°	based on Va angle, see "Vector" menu
Power Energy	Forward active power	0.000 – 999.99 [MW]	±0.5%	+P
	Reverse active power	0.000 – 999.99 [MW]	±0.5%	-P
	Forward reactive power	0.000 – 999.99 [MVar]	±0.5%	+Q, inductive load(L-load)
	Reverse reactive power	0.000 – 999.99 [MVar]	±0.5%	-Q, conductive load(C-load)
	Forward active energy	0.000 – 9999.99 [MWh]	±0.5%	+Ep
	Reverse active energy	0.000 – 9999.99 [MWh]	±0.5%	-Ep
	Forward reactive energy	0.000 – 9999.99 [MVarh]	±0.5%	+Eq
	Reverse reactive energy	0.000 – 9999.99 [MVarh]	±0.5%	-Eq
	Power factor	-1.00 – +1.00 [F/R]	±1.0%	+: LEAD, -: LAG F: Forward, R: Reverse
	Frequency	45.000 – 65.000 [Hz]	±0.005Hz	based on Va
Harmonics	Voltage	0.00 – 100.00 [%]	-	2nd–15th, TDD
	Current	0.00 – 100.00 [%]	-	2nd–15th, THD, TDD, K-factor
Demand	Current demand	0.000 – 999.99 [kA]	-	Demand peak
	Active power demand	0.000 – 9999.99 [MW]	-	Demand peak
	Reactive power demand	0.000 – 999.99 [MVar]	-	Demand peak

※ 1) In a delta connection, the voltage applied to the PT is indicated as the line-to-line voltage.

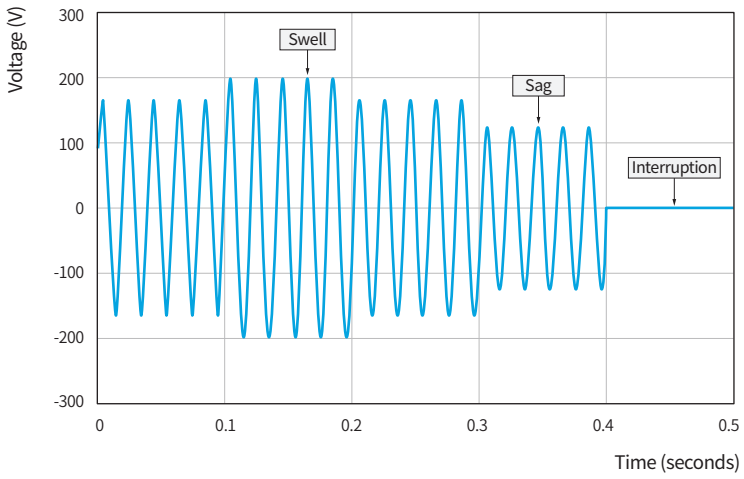
Features

Item		Specification	Description
CB Failure (CBF)	Delay time setting	50 – 10,000ms	It monitors delay time between the breaker control(ON/OFF) command and the feedback signal from the breaker. If it exceeds the set time, CBF alarm is activated.
	Output	User-defined LED, DO	
PT Fuse Failure (PTF)	Delay time setting	50 – 10,000ms	It monitors the state of PT circuit fuse and operates when all 3phase currents are between 10% and 100% of the rated value, while the measured voltage is zero.
	Output	User-defined LED, DO	
Power Fail	Operating voltage	Under 70VDC in control power	If the external control power falls below the reference voltage, system error has occurred, and the relay restricts the protection and DO output.
	Drop-out voltage	Over 88VDC in control power	
	Output	Sys.Err LED, User-defined LED, DO	
Counter	Trip Counter	0 – 65,535 time	Accumulated counts for CB open and CB close
	DO Counter	0 – 65,535 time	Accumulated counts for DO01–DO26 close
	CB on Time	0 – 4,294,697,295 hour	Total operating time during the breaker is closed
	Sys. Run Time	0 – 4,294,697,295 hour	Accumulated operating time record

Features

Power Quality Monitoring

The device measures and monitors Sag, Swell, and Interruption events, and records up to 256 PQ events along with detailed data such as PQ peak values and event duration.



Sag (Voltage Dip)

A sag is defined as a condition in which the RMS voltage remains between 10% and 90% of the rated voltage for a specified duration. (More specifically, Instantaneous sag: 0.5 to 30 cycles, Momentary sag: 30 cycles to 3 seconds, Temporary sag: 3 seconds to 1 minute)

Sags are typically caused by natural phenomena such as lightning, or by system ground faults or short-circuit faults, during which the voltage remains depressed until the fault is cleared.

Such events may lead to motor speed fluctuations or stalling, breaker malfunction, failures of electronic control devices, and loss of synchronism in motors and generators.

Swell (Voltage Rise)

A swell is defined as a condition in which the RMS voltage remains at or above 110% of the rated voltage, up to a specified upper limit, for a certain duration.

(More specifically, Instantaneous swell, 110% to 180% for 0.5 to 30 cycles, Momentary swell: 110% to 140% for 30 cycles to 3 seconds, Temporary swell: 110% to 120% for 3 seconds to 1 minute)

Swells may occur due to system ground faults, switching of large loads, or switching of large capacitor banks. Their effects may include motor speed variation or stoppage, breaker malfunction, failure of electronic controls, and loss of synchronism in motors and generators.

Interruption

An interruption is defined as a condition in which the RMS voltage remains at or below 10% of the rated voltage for a specified duration. (More specifically, Momentary interruption: 0.5 cycles to 3 seconds, Temporary interruption: 3 seconds to 1 minute)

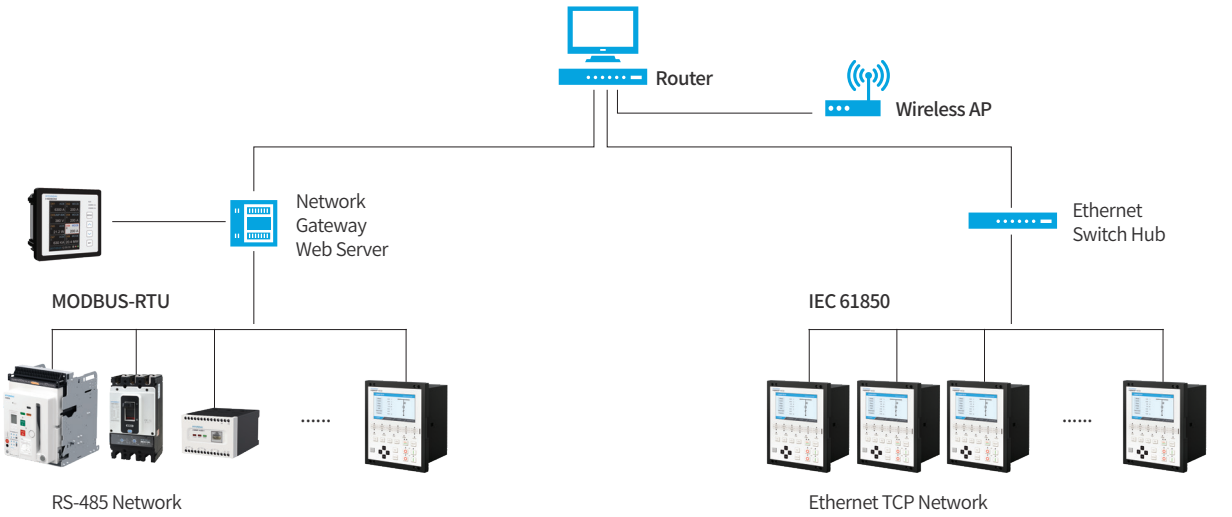
Interruptions share the same causes and effects as voltage sags.

Technical Data

Features

High-Speed Serial Communication

High-speed network integration is supported by selecting either the LAN or Fiber communication option through the Ethernet TCP communication module. The device complies with the IEC 61850 international communication standard for power systems and supports the Edition 2.0 stack.



High-Speed Redundant Communication - HSR (High-availability Seamless Redundancy) / PRP (Parallel Redundancy Protocol)

In response to smart-grid power applications, the device supports the HSR/PRP protocols defined in the international standard IEC 62439-3 for network fault-tolerance. These protocols ensure zero network frame loss and provide zero recovery time during network failures.

HSR's ring-type topology supports reduced infrastructure costs for redundant network construction.

Compared to existing ring-topology solutions such as RSTP – where reconfiguration may require several hundred milliseconds – HSR allows seamless system switchover without latency.

By implementing HSR and PRP solutions, users can build an economical and highly reliable network architecture.

Protection Relay

25, Synchronous Check Relay(SYNC)

When reconnecting two power systems that have been separated—typically due to maintenance or other operational requirements—it is essential to verify their synchronization prior to closing the circuit breaker.

The synchronous check relay compares the voltage magnitude, frequency, and phase angle between the two isolated systems. By evaluating whether these parameters fall within the predefined acceptable limits, the relay determines whether synchronization conditions are satisfied.

27, Undervoltage Relay (UVR)

The undervoltage relay operates when the phase voltage of the system falls below the preset threshold. This protection functions with a definite-time characteristic, initiating operation after an adjustable delay. The delay time can be configured within the range of 50ms to 60s.

The undervoltage relay can be used for various applications, including undervoltage protection, bus de-energization detection, load transfer, capacitor protection, and interlocking functions.

To prevent unintended operation caused by transient voltage dips during system energization, a UV Block function may be enabled. This function activates the undervoltage protection only after normal system voltage has been restored. To avoid misoperation due to an open fuse on the PT secondary circuit, it is recommended to use the PT Fuse Failure (PTF) function in coordination with the undervoltage relay.

27R, Residual Undervoltage Relay (UVR-X)

The residual undervoltage relay operates when the voltage of a specific phase in another bus system falls below the predetermined setting. This functions with a definite-time characteristic and initiates operation after an adjustable time delay.

The delay time can be set within a range of 100ms to 60s.

To prevent unintended operation caused by transient undervoltage during system energization, a UV Block function can be enabled. This feature activates the residual undervoltage protection only after normal system voltage has been restored.

32P, Active Power Relay (APR)

The active power relay is used in power systems with distributed generation to detect reverse power flow toward the utility grid. This protection operates when the measured active power exceeds the preset threshold value.

Depending on the direction of power flow, the relay can be configured for Forward or Reverse operation.

Both three-phase active power (Pset>Pa+Pb+Pc) and single-phase active power (Pset>Pa or Pb or Pc) can be independently monitored and detected. The operating characteristic follows a definite-time delay, which is adjustable from 100ms to 60s.

Technical Data

Protection Relay

32Q, Reactive Power Relay (RPR)

The reactive power relay is used in power systems with distributed generation where fault detection by overcurrent or ground-overcurrent relays may be insufficient due to the limited short-circuit capacity of the distributed source. Under such conditions, the relay measures reactive power using the fault-side current and the healthy-phase line-to-line voltage. When the measured reactive power exceeds the preset threshold, the relay operates.

Depending on the direction of reactive power flow, the relay can be configured for :

- Forward (Inductive Load)
- Reverse (Capacitive Load)Reactive power detection is based solely on single-phase reactive power(Qset>Qa or Qb or Qc). The relay uses a definite-time operating characteristic, with an adjustable delay ranging from 100ms to 60s.

37, Undercurrent Relay (UCR)

The undercurrent relay operates when the phase current of the system falls below the preset threshold.

This protection remains inactive when the circuit breaker is in the open position or before motor energization.

It operates with a definite-time characteristic, and the time delay can be adjusted within the range of 50ms to 60s.

37P, Under Active Power Relay (UPR)

The under active power relay detects conditions in which the generator output decreases due to line faults or generator malfunctions in systems incorporating distributed generation. This protection operates when the measured active power falls below the preset threshold value.

The relay supports detection in the Forward direction only. Both three-phase active power (Pset<Pa+Pb+Pc) and single-phase active power (Pset<Pa or Pb or Pc) can be individually monitored.

The operating characteristic is definite-time, and the time delay is adjustable from 100ms to 60s.

46, Negative Sequence Overcurrent Relay (NSOCR)

The negative sequence overcurrent relay calculates the negative-sequence current (I2) from the three-phase currents and detects abnormal conditions such as phase reversal, phase loss, and unbalanced load conditions. The relay operates after a time delay when the negative-sequence component exceeds the preset threshold.

Two operating characteristics are supported :

- Definite-time operation, where the relay trips after a fixed delay once the threshold is exceeded.
- Inverse-time operation, where the operating time decreases as the magnitude of the negative-sequence current increases. The inverse-time characteristic curves supported by the HGMAP-BCS include :
- IEC Standard: Standard Inverse (SI), Very Inverse (VI), Extremely Inverse (EI), Long-time Inverse (LI)
- IEEE Standard: Moderately Inverse (MI), Very Inverse (VI), Extremely Inverse (EI)
- KEPCO Guidelines : Standard Inverse (SI), Very Inverse (VI)

47N, Negative Sequence Overvoltage Relay (NSOVR)

The negative sequence overvoltage relay calculates the negative-sequence voltage (V2) from the three-phase voltages and detects abnormal conditions such as phase reversal, phase loss, and voltage unbalance in motor and power system applications.
This protection operates with a definite-time characteristic, and the operating delay is adjustable within therange of 50ms to 60s.

47P, Phase Open Relay (POR)

The voltage unbalance relay computes the voltage unbalance ratio (Vunbal) from the three-phase voltages and detects conditions such as voltage unbalance and phase loss.
This protection operates with a definite-time characteristic, and the operating delay can be set within the range of 50ms to 60s. The formula used in the HGMAP-BCS for calculating the voltage unbalance ratio is as follows :

$$V_{unbal}\,(\%) = \left(\frac{V_{max} - V_{min}}{V_{avg}} \right) \times 100$$

Vunbal : Voltage unbalance ratio

Vmin : The lowest voltage among the three phase voltages

Vmax : The highest voltage among the three phase voltages

Vavg : The average value of the three phase voltages

48/51LR, Stall/Lock Relay

A stall condition occurs when the motor, during normal operation, becomes unable to maintain its rotational speed because the load torque exceeds the motor torque, resulting in significant speed reduction or complete stoppage.
A lock condition refers to a failure of the motor to start due to excessive load torque during the starting period.
The Stall/Lock relay monitors the motor load current to determine whether a Stall or Lock condition has occurred.
The Stall operates with a definite-time characteristic, initiating operation after a fixed delay once the condition is detected.
The Lock supports both definite-time and inverse-time characteristics.
The inverse-time characteristic curves conform to IEC standards and include:
- Very Inverse (VI)
- Extremely Inverse (EI)

49, Thermal Relay (THR)

The thermal relay calculates the accumulated thermal energy based on the motor operating current and determines thermal overload when the calculated thermal capacity exceeds the preset threshold. The relay operates with an inverse-time characteristic, in which the operating time decreases as the accumulated thermal value approaches or exceeds the setting, and increases when the thermal load is lower.
The thermal relay in the HGMAP-BCS incorporates inverse-time characteristic curves defined in IEC 60255-8, including both the Hot Curve and Cold Curve. The Cold Curve is based on thermal effects and the thermal time constant under cold-start conditions, while the Hot Curve additionally accounts for previously accumulated thermal energy, representing the behavior of the motor under warm-start conditions.

Technical Data

Protection Relay

50/51, Overcurrent Relay (OCR)

The overcurrent relay detects excessive current flowing in each phase of the system due to short-circuit faults or overload conditions. It consists of two characteristics.

- Instantaneous overcurrent (50), which operates without intentional delay
- Time-overcurrent (51), which operates after a preset time delay. The instantaneous (50) is designed to operate within 50ms to 60s when a very high fault current occurs, such as a short-circuit on the load-side feeder, minimizing the spread of the fault. The time-overcurrent (51) supports both definite-time and inverse-time characteristics.
- Definite-time characteristic : When the current exceeds the preset threshold, the relay operates after a fixed time delay, regardless of the magnitude of the overcurrent. The delay time is adjustable within 50ms to 60s.
- Inverse-time characteristic: The operating time decreases as the magnitude of the overcurrent increases; higher fault currents result in shorter operating times, while lower fault currents extend the operating time. The inverse-time curves supported by the HGMAP-BCS include:
- IEC Standard : Standard Inverse (SI), Very Inverse (VI), Extremely Inverse (EI), Long-time Inverse (LI)
- IEEE Standard : Moderately Inverse (MI), Very Inverse (VI), Extremely Inverse (EI)
- KEPCO Guidelines : Standard Inverse (SI), Very Inverse (VI)

50/51N, Ground Overcurrent Relay (OCGR)

The ground overcurrent relay detects zero-sequence current exceeding the preset threshold when a ground fault occurs in a grounded system. Similar to the phase overcurrent relay, the OCGR consists of two characteristics :

- Instantaneous ground overcurrent (50N) : operates without intentional delay
- Time-over ground overcurrent (51N) : operates after a preset time delay

To prevent maloperation caused by temporary inrush or irregular currents during breaker closing or motor starting, the HGMAP-BCS includes an OCGR Block function, which disables the ground overcurrent protection for a user-defined duration after relay energization. The inverse-time characteristic curves supported by the HGMAP-BCS include:

- IEC Standard : Standard Inverse (SI), Very Inverse (VI), Extremely Inverse (EI), Long-time Inverse (LI)
- IEEE Standard : Moderately Inverse (MI), Very Inverse (VI), Extremely Inverse (EI)
- KEPCO Guidelines : Standard Inverse (SI), Very Inverse (VI)

59, Overvoltage Relay (OVR)

The overvoltage relay detects excessive voltage on each phase of the system when the measured voltage exceeds the preset threshold. Such conditions may occur due to voltage unbalance, generator voltage-regulation failure, or sudden loss of major loads. This protection operates with either definite-time or inverse-time characteristics. For definite-time operation, the relay trips after a fixed delay once the overvoltage exceeds the setpoint. The delay time is adjustable within the range of 50ms to 60s.

For inverse-time operation, the relay supports a wide range of characteristic curves, including:

- IEC Standard : Standard Inverse (SI), Very Inverse (VI), Extremely Inverse (EI), Long-time Inverse (LI)
- IEEE Standard : Moderately Inverse (MI), Very Inverse (VI), Extremely Inverse (EI)
- KEPCO Guidelines : Standard Inverse (SI), Very Inverse (VI)

64, Overvoltage Ground Relay (OVGR)

The overvoltage ground relay detects excessive zero-sequence voltage that appears in a grounded system when a ground fault occurs and the measured value exceeds the preset threshold. This protection can operate with either definite-time or inverse-time characteristics. For definite-time operation, the relay trips after a fixed delay, which can be set within the range of 40ms to 60s. For inverse-time operation, the relay supports the following characteristic curves:

- IEC Standard : Standard Inverse (SI), Very Inverse (VI), Extremely Inverse (EI), Long-time Inverse (LI)
- IEEE Standard : Moderately Inverse (MI), Very Inverse (VI), Extremely Inverse (EI)
- KEPCO Guidelines : Standard Inverse (SI), Very Inverse (VI) Because ground-fault detection using zero-sequence overvoltage alone can be unreliable, the OVGR should be used primarily as an alarm or auxiliary indicator. For accurate ground-fault protection, it is recommended to use the OVGR in conjunction with a Directional Ground Relay (DOCGR) or a Selected Ground Overcurrent Relay (SGR)

66, Notching Relay (NCH)

When a motor trips due to malfunction or overload, repeated attempts to restart the motor without addressing the root cause can lead to severe thermal stress and significant damage to the system due to high inrush current. The Notching Relay prevents such repeated and rapid restart attempts by monitoring motor start frequency and conditions.
The Notching Relay in the HGMAP-BCS supports four configurable functions:

1. Limiting the number of allowable motor starts within a preset time window
2. Restricting restart attempts when the interval between successive starts is shorter than the defined minimum time
3. Applying a mandatory time delay before allowing a restart after the motor has stopped
4. Restricting motor starting based on the residual thermal capacity calculated by the thermal relay These functions help protect the motor and associated equipment from excessive thermal loading and enhance overall system reliability.

67P, Directional Overcurrent Relay (DOCR)

The directional overcurrent relay detects excessive current exceeding the preset threshold in a three-phase system and consists of instantaneous and time-overcurrent. Because the relay incorporates directional characteristics, it trips only for faults occurring within the designated protection zone, providing selective fault isolation. Determining direction based solely on fault-phase voltage and current is difficult. Therefore, the directional relay uses line-to-line voltage as a reference. For example, when detecting an A-phase fault, the relay determines the fault current direction by referencing the phase angle of the BC line-to-line voltage, using predefined characteristic and operating angle ranges. During three-phase faults, the system voltage may collapse completely, making directional decision impossible. To address this, the HGMAP-BCS provides a Voltage Loss Memory function, which uses the stored pre-fault voltage phase (captured up to one second before the fault) as a reference for directional determination.
The relay supports the following operating characteristics :

* Instantaneous operation with a delay adjustable from 50ms to 60s
* Definite-time delayed operation
* Inverse-time operation, including :
  - IEC Standard : Standard Inverse (SI), Long-time Inverse (LI), Very Inverse (VI), Extremely Inverse (EI)
  - IEEE Standard : Moderately Inverse (MI), Very Inverse (VI), Extremely Inverse (EI)
  - KEPCO guidelines : Standard Inverse (SI), Very Inverse (VI)

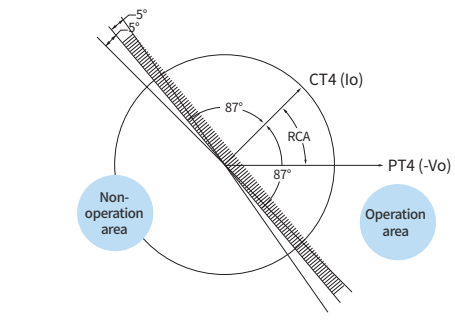
67N, Directional Ground Overcurrent Relay (DOCGR)

The directional ground overcurrent relay detects ground faults in grounded power systems by measuring the magnitude and phase angle of the zero-sequence voltage and zero-sequence current. The relay operates only when a ground fault occurs within the designated load-side bus section, ensuring selective fault isolation.
The DOCGR supports the following operating characteristics :

* Instantaneous operation, with an adjustable delay range of 50ms to 60s
* Definite-time operation, where the tripping delay can also be set from 50ms to 60s
* Inverse-time operation, supporting the following characteristic curves :
  - IEC Standard : Standard Inverse (SI), Very Inverse (VI), Extremely Inverse (EI), Long-time Inverse (LI)
  - IEEE Standard : Moderately Inverse (MI), Very Inverse (VI), Extremely Inverse (EI)
  - KEPCO Guidelines : Standard Inverse (SI), Very Inverse (VI)

67G, Selective Ground Overcurrent Relay (SGR)

The selected ground overcurrent relay operates in ungrounded power systems by detecting the magnitude and phase angle of the zero-sequence current and zero-sequence voltage measured through the built-in ZCT and PT when a ground fault occurs.
This relay operates only when a ground fault originates within the load-side bus system, enabling selective fault detection.
The relay uses a definite-time characteristic (67G), and the operating delay can be adjusted within the range of 50ms to 60s.



Technical Data

Protection Relay

78V, Voltage Vector Shift Relay (VVSR)

When both the utility grid and distributed generation (DG) supply power simultaneously, the DG must be disconnected if the grid supply is lost due to outages or system faults. Failure to disconnect the DG may result in severe transient currents when the grid is restored and the DG is not synchronized with the utility source.

The Voltage Vector Shift Relay detects sudden phase displacement of the DG voltage and operates based on the phase angle shift of the A-phase voltage. This protection provides instantaneous operation, responding within 40ms.

To prevent unintended operation caused by transient undervoltage during system energization, a UV Block function may be enabled. This feature activates the vector-shift protection only after normal system voltage has been restored.

79, Reclosing Relay (RCLS)

The reclosing relay automatically re-energizes a transmission line after it has been opened by protective relay operation following a system fault. Most transmission-line faults caused by lightning are transient in nature; once the arc is naturally extinguished, the insulation strength typically recovers, allowing successful re-energization. Automatic reclosing minimizes outage duration and improves system reliability.

The relay allows configuration of reclosing attempts, reclaim time, dead time, and reset time, depending on the system conditions into which the line is being re-energized.

81O/81U/81R, Frequency Relay

In systems where two power sources feed a tie line, or where distributed generation (DG) becomes isolated and supplies only a portion of the load in an islanding condition, the system frequency may deviate significantly. Under such circumstances, for stable system operation and the protection of connected equipment, the DG must be disconnected when the frequency departs from normal operating limits.

The frequency relay detects conditions where the system frequency becomes higher (81O) or lower (81U) than the rated value, or when the frequency deviates from any preset operating point. The Rate-of-Change-of-Frequency Relay (FROC, 81R) measures the change in frequency over time by comparing the frequency difference between successive cycles and converting it into a rate value (Hz/s). The relay operates when this rate exceeds the preset threshold.

Frequency measurement is based on the A-phase voltage, and the relay operates with a definite-time characteristic. The operating delay can be set from 100ms up to 60s, while the 81R (ROCOF) supports an adjustable delay from 200ms up to 60s.

87/87N, Differential Current Relay (DFR)

The differential current relay is a protection that receives current inputs from the primary and secondary circuits of equipment such as transformers and generators. It is designed to remain stable and restrain operation during normal conditions and external faults, while providing selective tripping only for internal faults such as line-to-line faults, layer short circuits, and ground faults.

The operating principle is based on the relationship between the restraining current (the scalar sum of the primary and secondary currents) and the differential current (the vector difference between the primary and secondary currents). The relay operates when the slope which is defined as the ratio between the differential current and the restraining current exceeds the preset threshold.

The DFR-87 relay operates on the per-phase percentage differential current of the primary and secondary circuits, whereas the DFR-87N relay operates on the percentage differential current derived from the transformer neutral current and the zero-sequence components of the three-phase currents. The operating time follows a definite-time characteristic and may be configured as instantaneous or with a time delay from 50ms to 60s.

Protection Relay

24, Overexcitation Relay (OEX)

During normal generator operation, the ratio of voltage to frequency remains stable, ensuring that the field current stays within its normal operating range. However, when an overexcitation condition occurs such as during sudden load rejection the generator may enter magnetic saturation, causing the field current to rise excessively. This can lead to generator overheating, potential insulation damage, and increased capacitive reactive power injection into the system, resulting in power factor distortion. The overexcitation relay detects such abnormal conditions by monitoring the voltage-to-frequency (V/Hz) ratio and operates when the measured value exceeds the preset limit. It can be set within a time range of 100ms to 60s.

40Q, Loss of Field Relay (LOF)

The loss of field relay detects short-circuits or ground faults within the generator field circuit.
When a loss-of-field condition occurs due to a fault in the excitation system, the field current decreases sharply, causing the generator to operate in a leading power factor mode and to absorb reactive power from the system. This adversely affects system power-factor control and overall voltage stability.
In addition, the reduction in magnetic coupling between the generator rotor and stator may cause the generator to accelerate toward overspeed conditions. This can lead to excessive leading currents in the stator windings, resulting in insulation breakdown of the field windings, excessive temperature rise, and potentially severe generator damage.
It calculates the impedance using the phase-to-phase voltages and currents, and determines a loss-of-excitation condition when the impedance variation(trajectory) enters a defined operating region. Considering factors such as excitation recovery time, the operating time can be set within a range of 100ms to 60s.

78S, Out-of-Step Tripping Relay (OST)

Out-of-step conditions may occur when the generator experiences a sudden change in load, causing it to deviate from its normal operating speed. In such conditions, the generator loses synchronism with the system voltage and frequency, resulting in rapid and unstable variations in phase angle. Once synchronism is lost, the generator can no longer maintain stable power delivery to the system.
The Out-of-Step Tripping Relay monitors the active power flow on the feeder side to detect these out-of-step conditions and initiate tripping when instability is identified.
It determines a loss of synchronism by monitoring the direction of active power resulting from unstable rotor angle variations. Considering factors such as synchronizing power, the number of counts and the operating time required for detection can be configured within a range of 100ms to 60s.

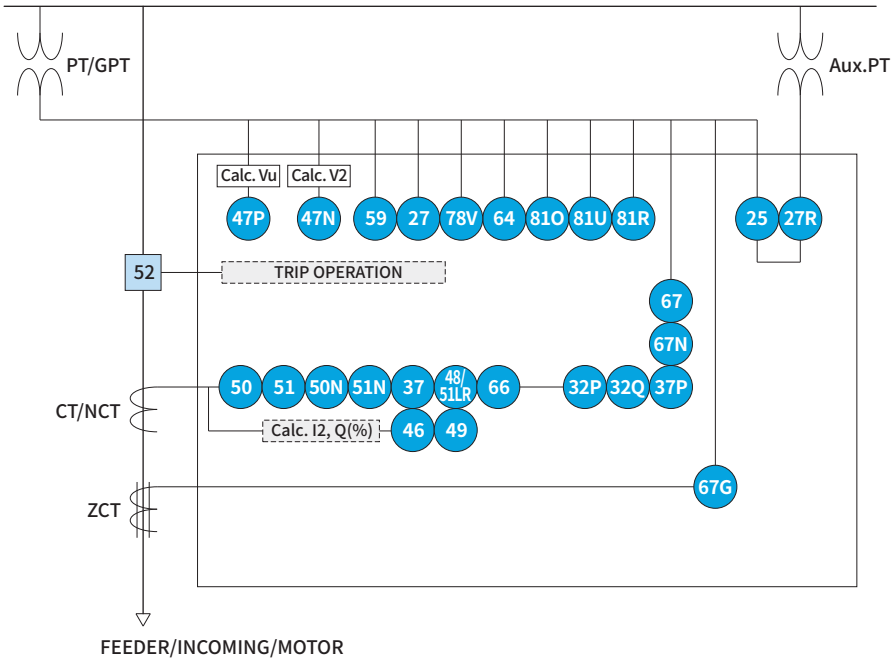
95I, Inrush Block Relay (INRS)

When power is initially applied, such as during breaker closing capacitors, transformers, motors, and other inductive loads, it may draw momentary inrush current due to their low initial impedance. Such inrush currents can be significantly higher than normal operating current and may cause maloperation of instantaneous overcurrent or differential protection.
The Inrush Block function detects these conditions and prevents unnecessary tripping. When enabled, the relay monitors harmonic content to accurately identify inrush current and applies this detection as an additional restraint condition for the associated protection.
It detects energization of transformer or load by monitoring the second-harmonic content contained in the transient overcurrent. Considering factors such as transformer capacity and load characteristics, the block time can be configured within a range of 40ms to 60s.

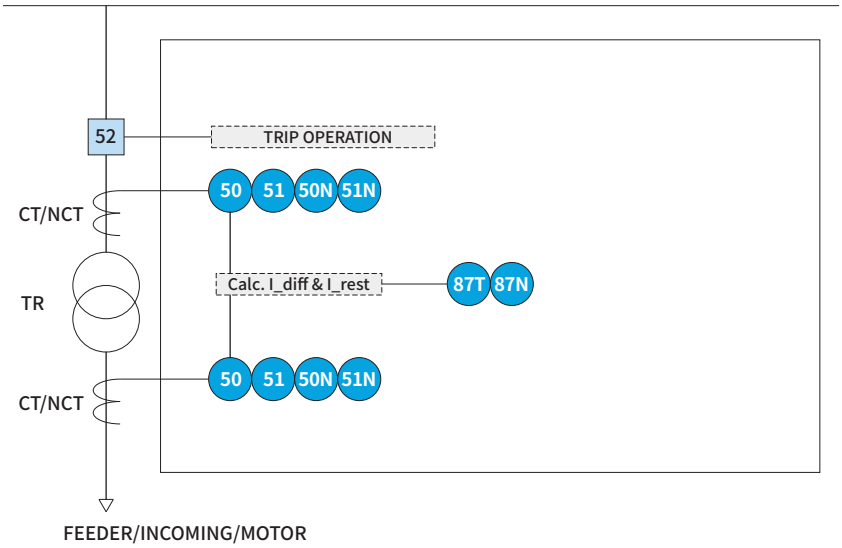
Technical Data

Function Block Diagram

Feeder/Incoming/Motor

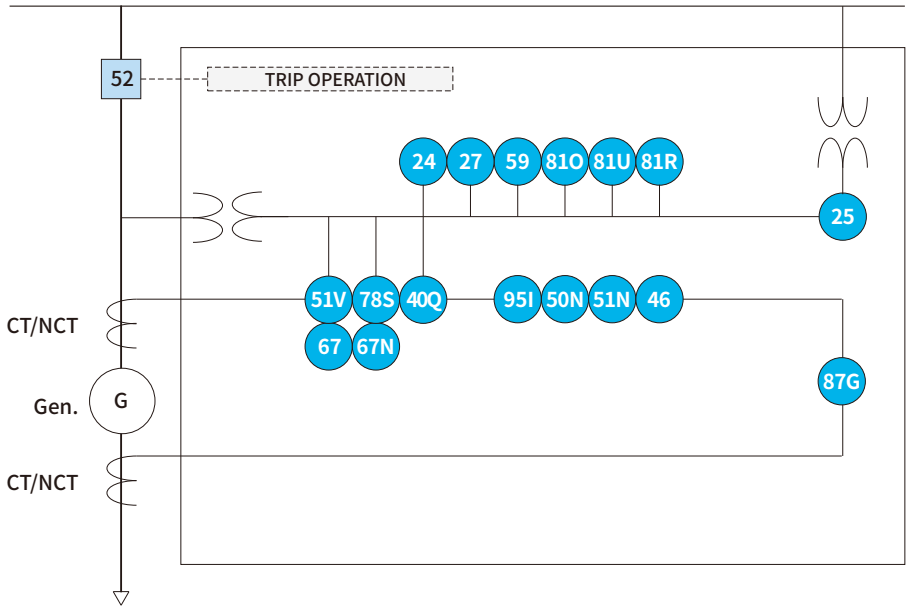


Transformer



Function Block Diagram

Generator



Technical Data

Protection Relay Setting

Relay	Max.Stage	Input Signal	Operation	Set Range	Characteristic	Operating Time	Note
OEX (24)	2	Va – Vc	dU/dF >>	OFF, 0.10 – 99.90%	Definite	0.10 – 60.00 s (0.01 step)	
			UV Block	0.50 – 0.90 Vn (0.01 step)			
SYNC (25)	1	Va, Vo	Voltage Diff.	2 – 80 V (1 step)	Definite	0.05 – 60.00 s (0.01 step)	
			Phase Diff.	5 – 100 (1 step)			
			Freq. Diff.	0.01 – 5.00 Hz (0.01 step)			
			Dead Voltage	OFF, 0.20 – 0.40 Vn (0.01 step)	-		
UVR (27)	2	Va – Vc	V <<	OFF, 0.20 – 0.90 Vn (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	
			Auto Reset : Automatic reset after power is restored				
			UV Block : Block operation if phase voltage is below 15V in PT secondary				
UVR-X (27R)	1	Vo	V <<	OFF, 0.20 – 0.90 Vn (0.01 step)	Definite	0.10 – 60.00 s (0.01 step)	
			Auto Reset : Automatic reset after power is restored				
			UV Block : Block operation if phase voltage is below 15V in PT secondary				
APR (32P)	2	Ia_P – Ic_P Va – Vc	P >>	OFF, 0.01 – 1.50 Pn (0.01 step)	Definite	0.10 – 60.00 s (0.01 step)	
			Mode	Forward 1, Forward 3, Reverse 1, Reverse 3			
RPR (32Q)	2	Ia_P – Ic_P Va – Vc	Q >>	OFF, 0.01 – 1.50 Qn (0.01 step)	Definite	0.10 – 60.00 s (0.01 step)	
			Mode	Forward 1, Reverse 1			
UCR (37)	2	Ia_P – Ic_P	I <<	OFF, 0.10 – 0.90 In (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	
UPR (37P)	2	Ia_P – Ic_P Va – Vc	P <<	OFF, 0.01 – 0.80 Pn (0.01 step)	Definite	0.10 – 60.00 s (0.01 step)	
			Mode	Forward 1, Forward 3			
LOF (40Q)	2	Ia_P – Ic_P Va – Vc	Diameter	0.10 – 99.99 Ohm (0.01 step)	Definite	0.10 – 60.00 s (0.01 step)	
			Offset	0.10 – 0.40 Von (0.01 step)			
			Q.Limit	OFF, 0.01 – 1.50 Pn (0.01 step)	Definite	0.10 – 60.00 s (0.01 step)	
			UV Block	0.50 – 0.90 Vn (0.01 step)			
NSOCR (46)	2	Calc. I2	I2 >	OFF, 0.10 – 1.00 In (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	DT, IEC(SI/VI/EI/LI), IEEE(MI/VI/EI),KEPCO(SI/VI)
					Inverse	0.05 – 10.00 (0.01 step)	
			UV Block	OFF, 0.20 – 1.00 Vn (0.01 step)			
NSOVR (47N)	2	Calc. V2	V2 >>	OFF, 0.05 – 1.00 Vn (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	
POR (47P)	2	Calc. Vu(%)	Vu >>	OFF, 5 – 100 % (1 step)	Definite	0.05 – 60.00 s (0.01 step)	%= (Vmax – Vmin) / Vavg
THR (49)	1	Calc. Q	Q >	OFF, 50 – 100 % (1 step)	Inverse		
					Time Delay	0.05 – 60.00 s (0.01 step)	
			Q.Limit	OFF, 0.10 – 4.00 FLC (1 step)			
OCR-I (50)	2	Ia_P – Ic_P Ia_S – Ic_S	I >>>	OFF, 0.5 – 20.0 In (0.1 step)	Instantaneous	0.05 – 60.00 s (0.01 step)	
			Inrush Block : Blocked during the period in which inrush current occurs at initial energization (related to INRS 95I)				Only for CT_pri
OCR-INV (51)	3	Ia_P – Ic_P Ia_S – Ic_S	I >	OFF, 0.10 – 10.00 In (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	DT, IEC(SI/VI/EI/LI), IEEE(MI/VI/EI),KEPCO(SI/VI)
					Inverse	0.05 – 10.00 (0.01 step)	
			Rest.V	OFF, 0.10 – 0.90 Vn (0.01 step)			ANSI 51V (Only for CT_pri)
OCGR-I (50N)	2	IN	I >>>	OFF, 0.10 – 8.00 In (0.01 step)	Instantaneous	0.05 – 60.00 s (0.01 step)	
			OCGR Block Time: Operation delay function to prevent malfunctions caused by inrush current after circuit breaker closing			0.05 – 60.00 s (0.01 step)	
OCGR-INV (51N)	2	IN	I >	OFF, 0.02 – 2.00 In (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	DT, IEC(SI/VI/EI/LI), IEEE(MI/VI/EI),KEPCO(SI/VI)
					Inverse	0.05 – 10.00 (0.01 step)	
			OCGR Block Time: Operation delay function to prevent malfunctions caused by inrush current after circuit breaker closing			0.05 – 60.00 s (0.01 step)	
STALL/ LOCK (48/51LR)	2	Ia_P – Ic_P	STALL >>	OFF, 0.50 – 10.00 FLC (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	
			LOCK >	OFF, 0.50 – 10.00 FLC (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	DT, IEC(VI/EI)
					Inverse	0.05 – 10.00 (0.01 step)	
OVR (59)	2	Va – Vc	V >	OFF, 0.80 – 1.60 Vn (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	DT, IEC(SI/VI/EI/LI), IEEE(MI/VI/EI),KEPCO(SI/VI)
					Inverse	0.05 – 10.00 (0.01 step)	
OVGR (64)	3	Vo	Vo >	OFF, 0.10 – 0.40 Von (0.01 step)	Definite	0.04 – 60.00 s (0.01 step)	DT, IEC(SI/VI/EI/LI), IEEE(MI/VI/EI),KEPCO(SI/VI)
					Inverse	0.05 – 10.00 (0.01 step)	
NCH (66)	1	Number of Start	Ns	OFF, 1 – 16 (1 step)	-	-	
		Operating Time	Tb	1 – 60 min (1 step)			
		Start/Stop Interval	Tss	OFF, 1 – 60 min (1 step)			
		Residual Heat	Qr	OFF, 10 – 80 % (1 step)			
DOCR-I (67-I)	2	Ia_P – Ic_P Va – Vc	I >>>	OFF, 0.5 – 20.0 In (0.1 step)	Instantaneous	0, 0.05–60.00 s(0.01 step)	
			RCA	0 – 359 ° (1 step)			
			OPA	± 60 – 87° (1 step)			
			Voltage loss block : If 3-phase voltage fault occurs, DOCR operation is blocked a second after the fault				

Relay	Max.Stage	Input Signal	Operation	Set Range	Characteristic	Operating Time	Note
DOCR-INV (67-D)	2	Ia_P – Ic_P Va – Vc	I >	OFF, 1.00 – 10.00 In (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	DT, IEC(SI/VI/EI/LI), IEEE(MI/VI/EI),KEPCO(SI/VI)
					Inverse	0.05 – 10.00 (0.01 step)	
			RCA	0 – 359° (1 step)			
			OPA	± 60 – 87° (1 step)			
			Voltage loss block : If 3-phase voltage fault occurs, DOCR operation is blocked a second after the fault				
DOCGR-I (67N-I)	2	Va – Vc, Vo IN	Io >>>	OFF, 0.10 – 8.00 In (0.01 step)	Instantaneous	0, 0.05–60.00 s(0.01 step)	
			Vo >>>	0.10 – 0.40 Von (0.01 step)			
			Vo Mode	GPT 3Vo, Calc 3Vo			
			RCA	0 – 359° (1 step)			Calc 3Vo :Vector sum of 3-phase voltages
			OPA	± 60 – 87° (1 step)			
DOCGR-INV (67N-D)	2	Va – Vc, Vo IN	Io >	OFF, 0.02 – 2.00 In (0.01 step)	Definite	0.05 – 60.00 s (0.01 step)	DT, IEC(SI/VI/EI/LI), IEEE(MI/VI/EI),KEPCO(SI/VI)
					Inverse	0.05 – 10.00 (0.01 step)	
			Vo >	0.10 – 0.40 Von (0.01 step)			
			Vo Mode	GPT 3Vo, Calc 3Vo			Calc 3Vo :Vector sum of 3-phase voltages
			RCA	0 – 359° (1 step)			
			OPA	± 60 – 87° (1 step)			
SGR (67G)	1	Vo, Io	Io >>	OFF, 0.6 – 4.0 Ion (0.1 step)	Definite	0.05 – 60.00 s (0.01 step)	
			Vo >>	OFF, 0.10 – 0.40 Von (0.01 step)			
			RCA	0 – 359° (1 step)			
			OPA	± 60 – 87° (1 step)			
OST (78S)	1	Ia_P – Ic_P Va – Vc	Rev.P Limit	0.01 – 10.00 Pn (0.01 step)	-	-	
			For.P Limit	0.01 – 20.00 Pn (0.01 step)			
			Count	1 – 100 (1 step)			
			Holding time	0.10 – 60.00 s (0.01 step)			
VVSR (78V)	1	Va – Vc	△δ	OFF, 2 – 30° (1 step)	Instantaneous	Within 50 ms	△δ : Change in angle
			UV Block	0.50 – 0.90 Vn (0.01 step)			
RCLS (79)	1	Reclosed Circuit Number		1 – 5 (1 step)	-	-	
		Prepare Time		0.20 – 60.00 s (0.01 step)			
		Reclaim Time		10.00 – 300.00 s (0.01 step)			
		Dead Time (1–5)		0.10 – 300.00 s (0.01 step)			
OFR (81O)	4	Va	Hz >>	OFF, 50.00 – 70.00 Hz (0.01 step)	Definite	0.10 – 60.00 s (0.01 step)	
			UV Block	0.50 – 0.90 Vn (0.01 step)			
UFR (81U)	4	Va	Hz <<	OFF, 50.00 – 70.00 Hz (0.01 step)	Definite	0.10 – 60.00 s (0.01 step)	
			UV Block	0.50 – 0.90 Vn (0.01 step)			
FROC (81R)	4	Va	Hz/s	OFF, 0.1 – 2.0 Hz/s (0.1 step)	Definite	0.20 – 60.00 s (0.01 step)	
			UV Block	0.50 – 0.90 Vn (0.01 step)			
DFR (87)	1	Ia_P – Ic_P Ia_S – Ic_S	I >>>	OFF, 1.0 – 20.0 In (0.1 step)	Instantaneous	Within 40 ms	
			I >>	OFF, 0.20 – 1.00 In (0.01 step)	Definite	0.04 – 60.00 s (0.01 step)	
			Knee point	1.0 – 15.0 In (0.1 step)			
			Slope1	15 – 100 % (1 step)			
			Slope2	15 – 100 % (1 step)			
			Inrush Block : Blocked during the period in which inrush current occurs at initial energization (related to INRS 95I)				Only for CT_pri
			Angle Comp. : Angle compensation function operates if TR Wiring is selected (protection for TR)				
DFRN (87N)	1	Ia_P – Ic_P IN	I >>	OFF, 0.10 – 1.00 In (0.01 step)	Definite	0.04 – 60.00 s (0.01 step)	
			Slope1	15 – 100 % (1 step)			
INRS (95I)	1	Ia_P – Ic_P	H2 >>	OFF, 10 – 90 % (1 step)	Definite	0.04 – 60.00 s (0.01 step)	
			OC Block	0.50 – 20.00 In (0.01 step), Blocked if the fundamental current exceeds the preset threshold			

※ Vn : PT secondary rating (110V or 110/√3V)
Von : GPT tertiary rating (110V or 190V)

In : CT secondary rating (5A)
Ion : ZCT secondary rating (1.5mA)

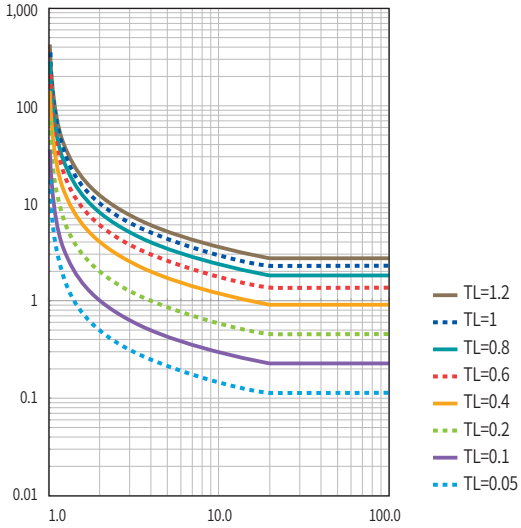
Motor Characteristic

Characteristic		Set Range	Note
Motor Start Time	MST	1.00 – 60.00 s (0.01 step)	
Full Load Current	FLC	0.20 – 2.00 In (0.01 step)	
Thermal Factor (S/F:Service Factor, O/C:Overload Const)	S/F	1.00 – 1.20 (0.05 step)	
	O/C	0.80 – 1.20 (0.05 step)	
Thermal Time Constant	Th	2 – 60 min (1 step)	
	Tc	2 – 60 min (1 step)	

Technical Data

Characteristic Curve

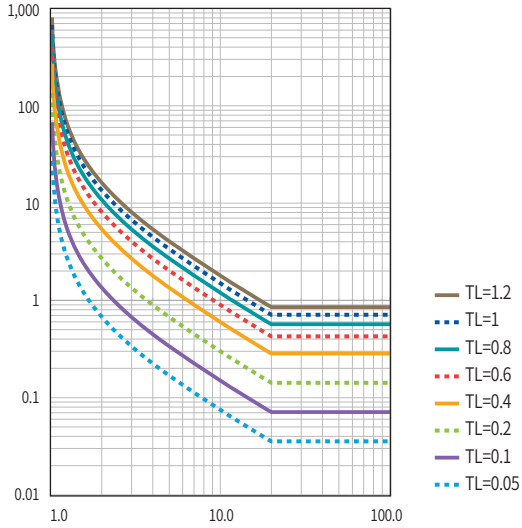
Inverse Curve : IEC_SI



$$T_{SI} = \left(\frac{0.14}{\left(\frac{I}{Is}\right)^{0.02} - 1} \right) \times T_L$$

unit : s
I : Fault current
Is : Setting current
T_L : Time Lever

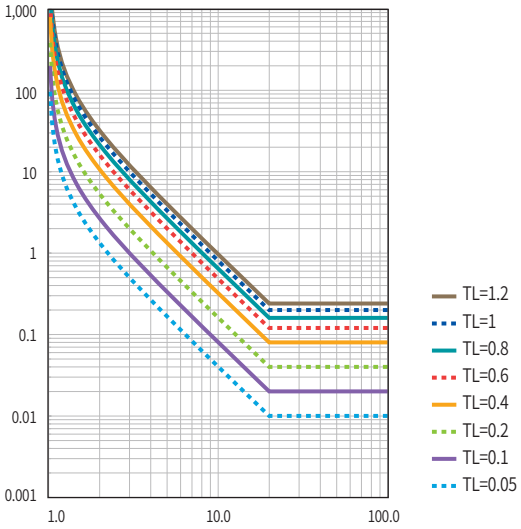
Inverse Curve : IEC_VI



$$T_{VI} = \left(\frac{13.5}{\left(\frac{I}{Is}\right) - 1} \right) \times T_L$$

unit : s
I : Fault current
Is : Setting current
T_L : Time Lever

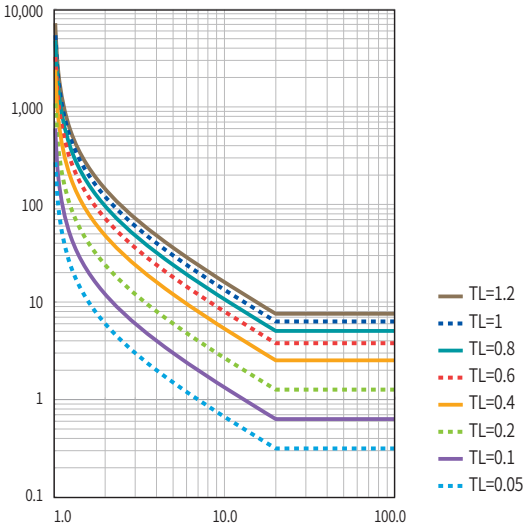
Inverse Curve : IEC_EI



$$T_{EI} = \left(\frac{80}{\left(\frac{I}{Is}\right)^{2} - 1} \right) \times T_L$$

unit : s
I : Fault current
Is : Setting current
T_L : Time Lever

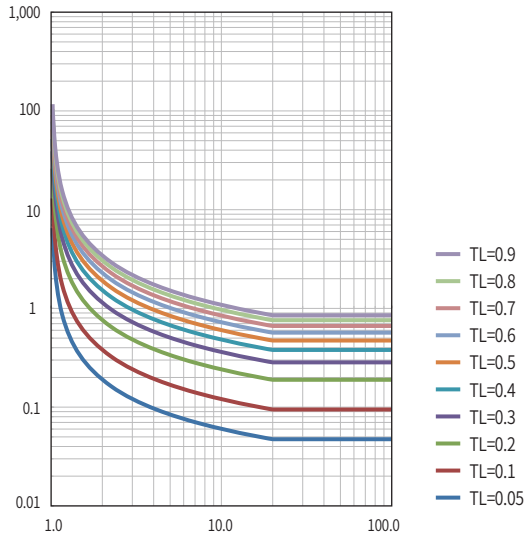
Inverse Curve : IEC_LI



$$T_{LI} = \left(\frac{120}{\left(\frac{I}{Is}\right) - 1} \right) \times T_L$$

unit : s
I : Fault current
Is : Setting current
T_L : Time Lever

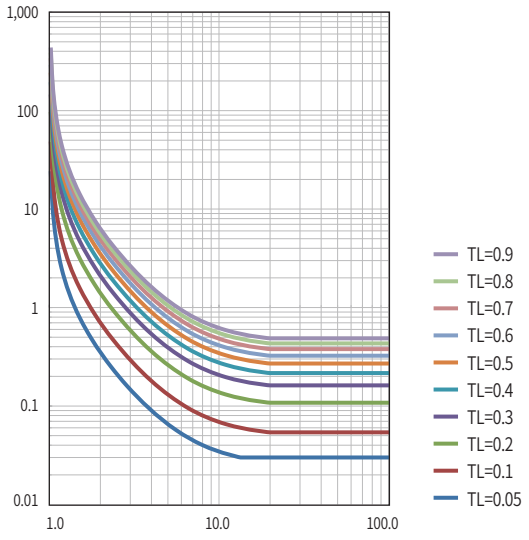
Inverse Curve : IEEE_MI



$$T_{MI} = \left(\frac{0.0515}{\left(\frac{I}{Is}\right)^{0.02} - 1} + 0.114 \right) \times T_L$$

unit : s
I : Fault current
Is : Setting current
T_L : Time Lever

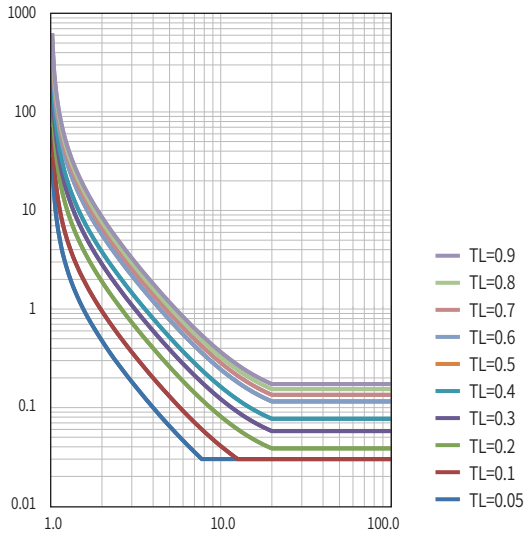
Inverse Curve : IEEE_VI



$$T_{VI} = \left(\frac{19.61}{\left(\frac{I}{Is}\right)^{2} - 1} + 0.491 \right) \times T_L$$

unit : s
I : Fault current
Is : Setting current
T_L : Time Lever

Inverse Curve : IEEE_EI



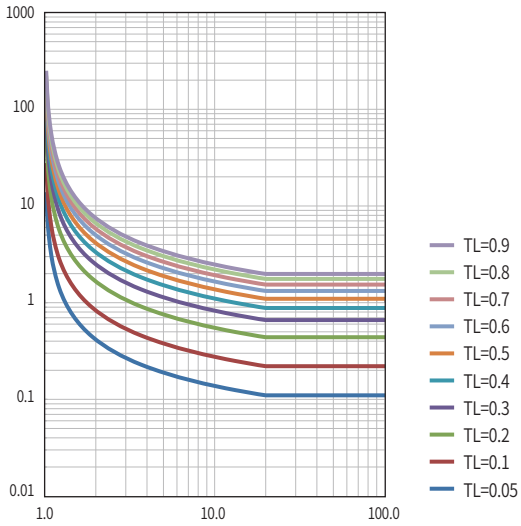
$$T_{EI} = \left(\frac{28.2}{\left(\frac{I}{Is}\right)^{2} - 1} + 0.1217 \right) \times T_L$$

unit : s
I : Fault current
Is : Setting current
T_L : Time Lever

Technical Data

Characteristic Curve

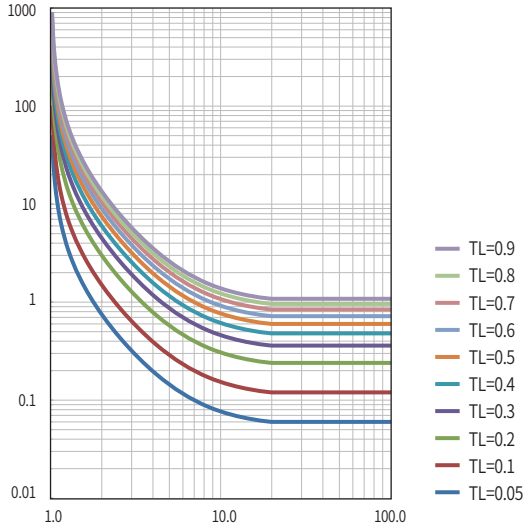
Inverse Curve : KEPCO_SI



$$T_{SI} = \left(\frac{0.11}{\left(\frac{I}{Is}\right)^{0.02} - 1} + 0.42 \right) \times T_L$$

unit : s
I : Fault current
Is : Setting current
T_L : Time Lever

Inverse Curve : KEPCO_VI

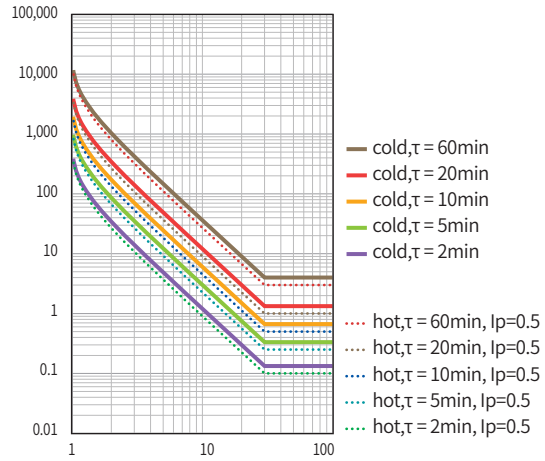


$$T_{VI} = \left(\frac{39.85}{\left(\frac{I}{Is}\right)^{1.95} - 1} + 1.084 \right) \times T_L$$

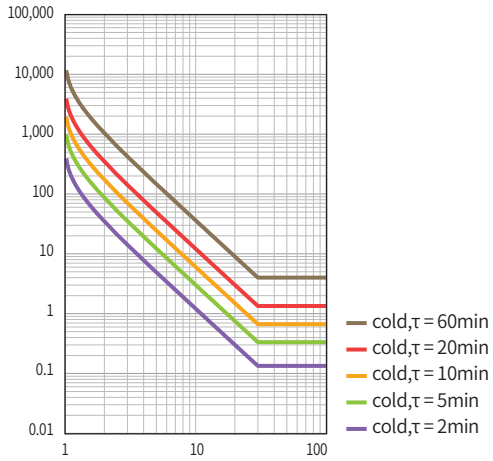
unit : s
I : Fault current
Is : Setting current
T_L : Time Lever

Thermal Curve Cold vs Hot

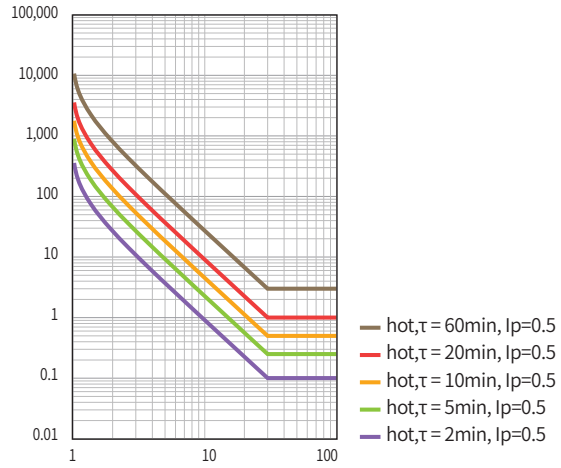
Cold vs Hot



Cold (FLC 0%)



Hot (FLC 50%)



$$t = \tau \times ln \frac{I^2 - Ip^2}{I^2 - (k \times I_B)^2}$$

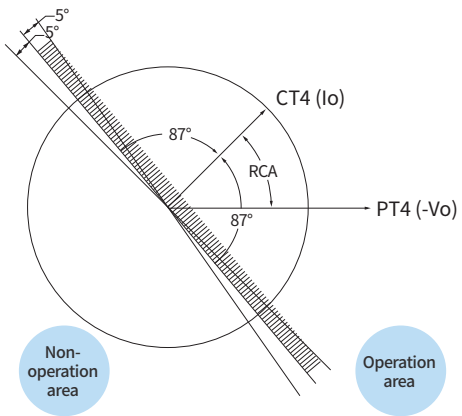
t : Trip time (s)
τ : Time constant
I_P : Pickup current
I_B : Rated current
k : Overload factor (FLC x S/F x O/C)
I : Fault current

- X axis : I/I_B (%)
- Y axis : Trip time (s)
- Cold condition $I_P = 0$

DOCGR/SGR Operating Area Characteristics

- The operating areas of the DOCGR and SGR are within ±87° from the preset Residual Characteristic Angle (RCA). However, the operating area of the DOCGR is within the designated operating area(60° to 87°) from the preset RCA.
- When connecting the PT4 terminal to the tertiary side of the GPT, ensure that it is wired to the "-Vo" terminal.
- The accuracy is within ±5° of the reference operating angle.

(Example) RCA set to 45°
- Operating area(based on -Vo) : -42° to 87°
- Phase difference : 82° to 92°, -47° to –37°



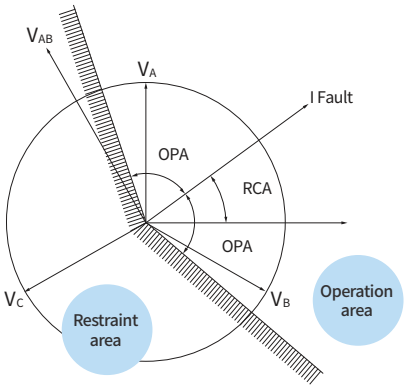
Technical Data

Characteristic Curve

DOCR Operation Area

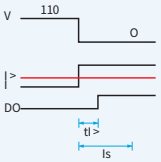
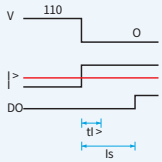
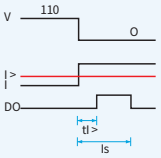
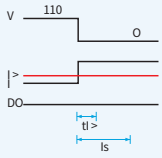
The operation area of the DOCR is within 60–87° based on the configured Residual Characteristic Angle (RCA).
For a phase-A fault, the DOCR determines the polarity by analyzing the phase difference between the phase-A current I_A and the line-to-line voltage V_{BC}.

Fault phase	Reference current	Reference voltage
A	I_A	$V_{BC} = V_B - V_C$
B	I_B	$V_{CA} = V_C - V_A$
C	I_C	$V_{AB} = V_A - V_B$



DOCR Voltage Loss Block Characteristics

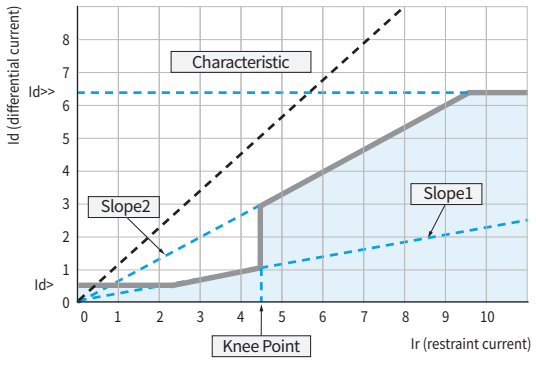
When a 3-phase short-circuit fault occurs, all phase voltages may drop below the reference level(11V), making it difficult for the DOCR to determine the direction of the fault current.
To address this issue, the relay memorizes the voltage waveform for one second prior to the fault and uses this stored voltage to determine the current direction.
Depending on whether the Voltage Loss Block function is enabled, the DOCR operation and the state of the DO output under each condition can be classified as shown.

Fault current	Within the operation area	Beyond the operation area
Voltage Loss Block : OFF		
Voltage Loss Block : ON		

DFR-87 Operation Area

The operation area of the DFR-87 corresponds to the region above the characteristic curve defined by the restraint current (Ir) on the x-axis and the configured values of slope1, slope2, and the knee point.
However, when the differential current (Id) exceeds the instantaneous setting (Id≫), the relay operates regardless of the restraint current.

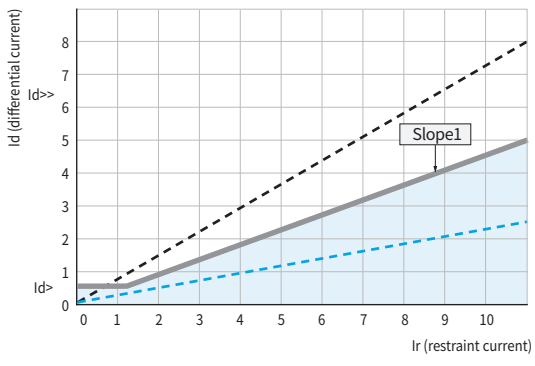
- Differential Current (Id) : Id = | I1 - I2 | (vector difference)
- Restraint Current (Ir) : Ir = |I1| + |I2| (scalar sum)



DFR-87N Operation Area

The operation area of the DFR-87N corresponds to the region above the characteristic curve defined by the restraint current (Ir) on the x-axis and the configured slope, differing from the DFR-87 characteristics.
In this case, both the differential current (Id) and the restraint current (Ir) operate based on the transformer neutral current and the zero-sequence current of the three-phase system.

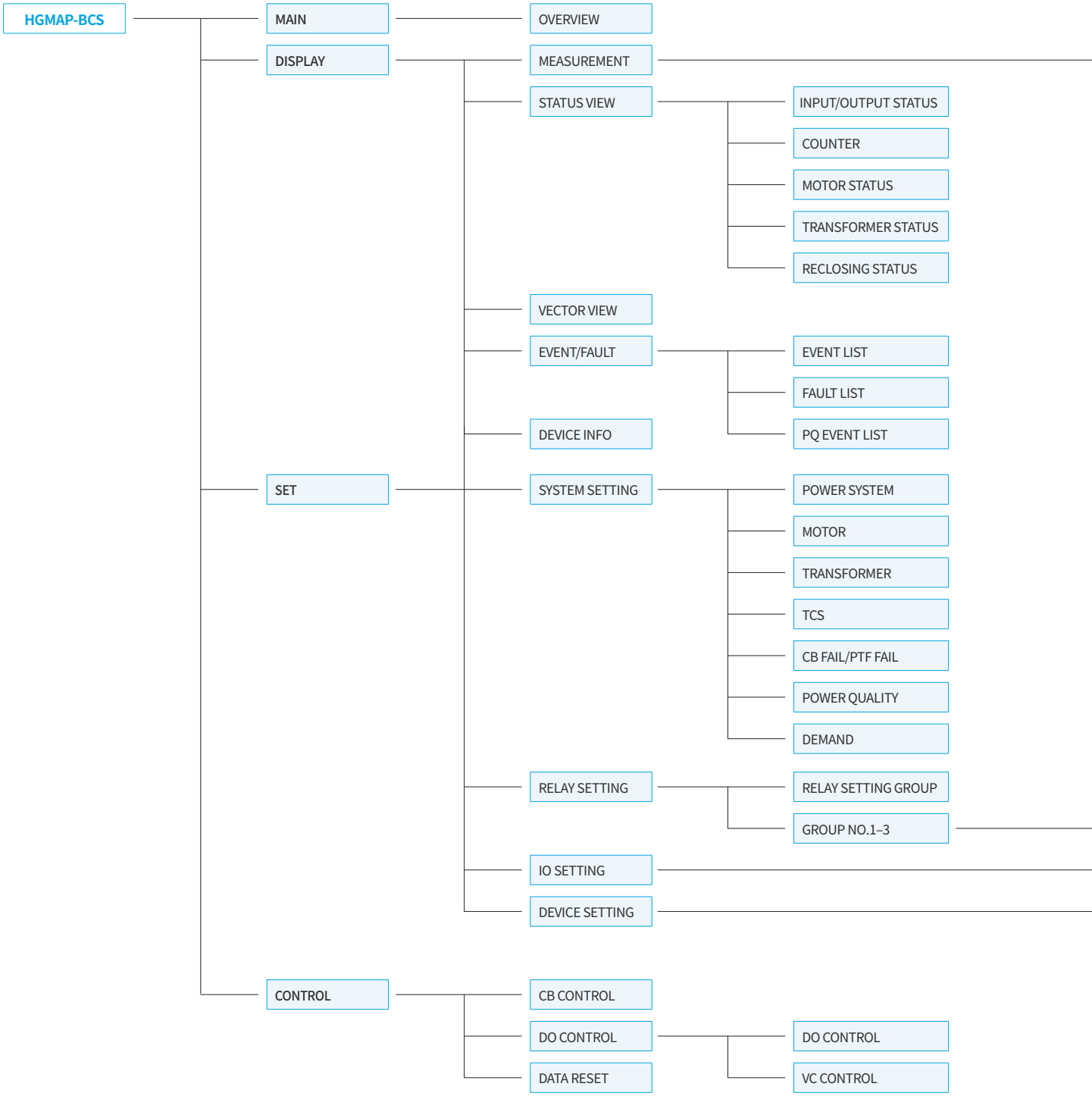
- Differential Current (Id) : Id = | 3I0 - Ig (vector difference)
- Restraint Current (Ir) : Ir = |3I0| + |Ig| (scalar sum)

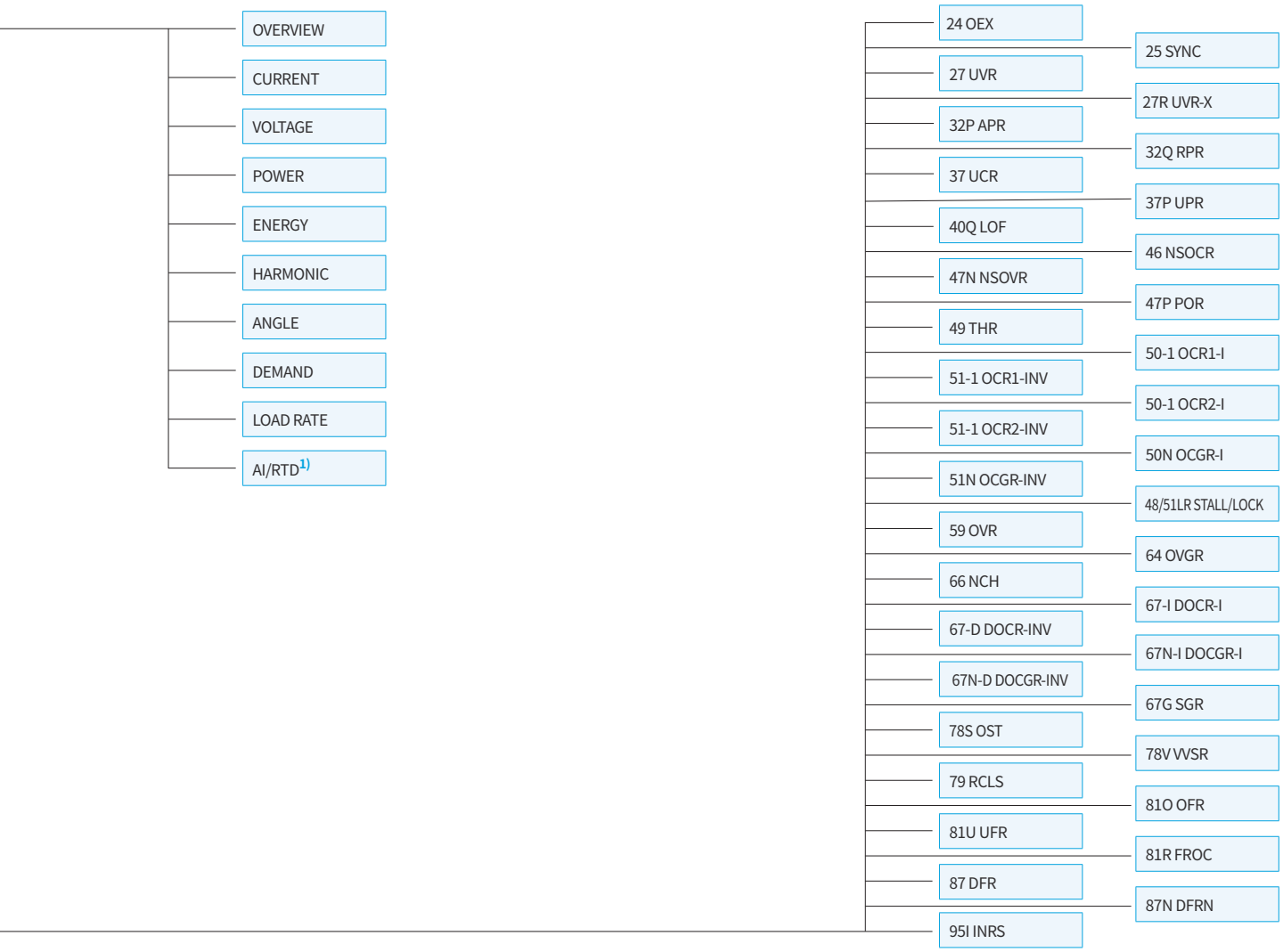


Technical Data

User Interface

HGMAP-BCS Menu Tree





※ 1) Optional menu
AI/RTD type will release in the first half of 2026, TCP type will release in 2026.

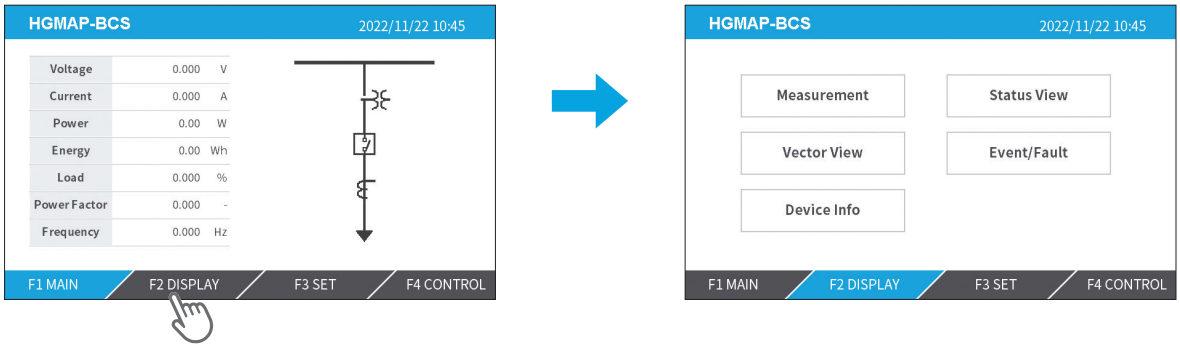
Technical Data

User Interface

Basic Operation Guide

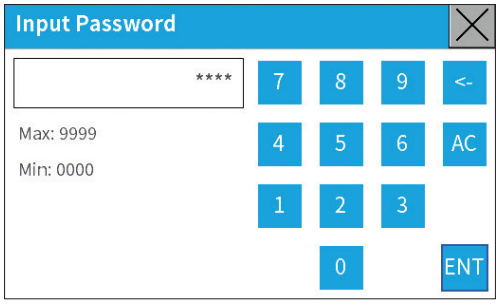
1. Select the Menu

When the SET button is touched on the main screen, the display moves to the settings menu.
Press the MAIN button again to return to the measurement screen.



2. Input Password

Protection relays operate in a locked state by default, and in this state, it is not possible to change relay settings or operate the breaker.
To make any changes, the user must enter the password configured for the device to release the lock state.
If an operation requiring lock release is attempted while the relay is locked, a password window will appear as shown below.



▶ **In the Password pop-up window, enter the four-digit password and press the ENT key.**

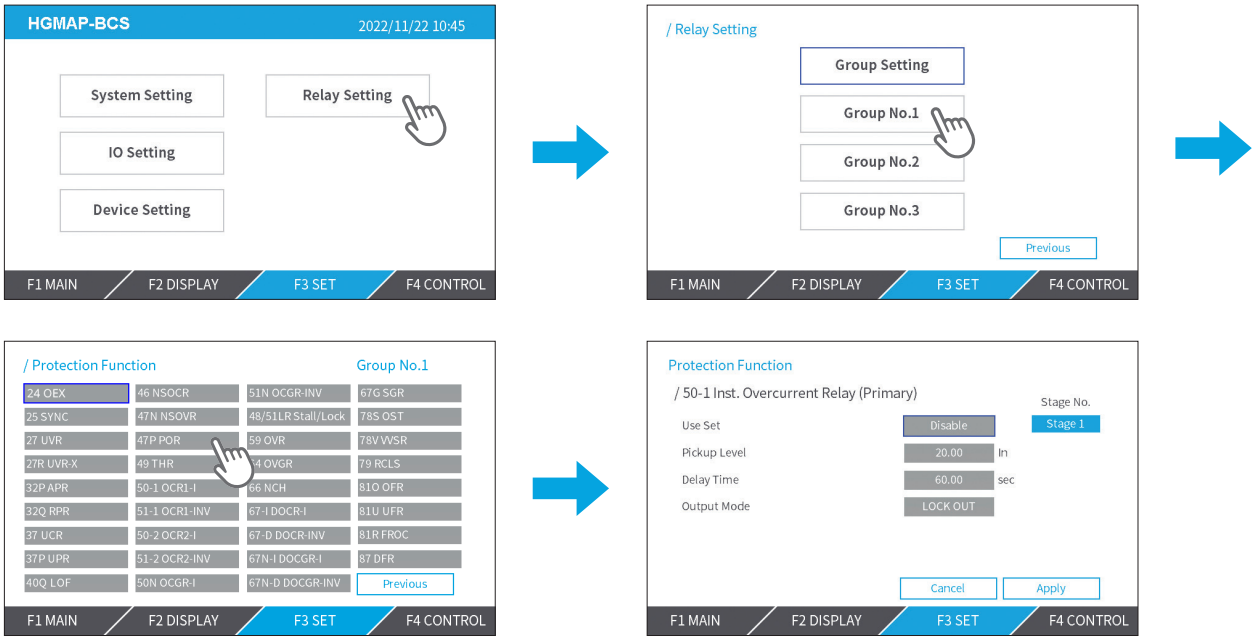
The default password set is 9999. For safe switchgear operation, it is recommended to change it to a unique password.
The password can be changed in the SET menu → DEVICE SETTING → PASSWORD section.

3. Relay Setting

The protection relay can be configured in the SET menu → Relay Setting page, where each protection relay can be individually set.

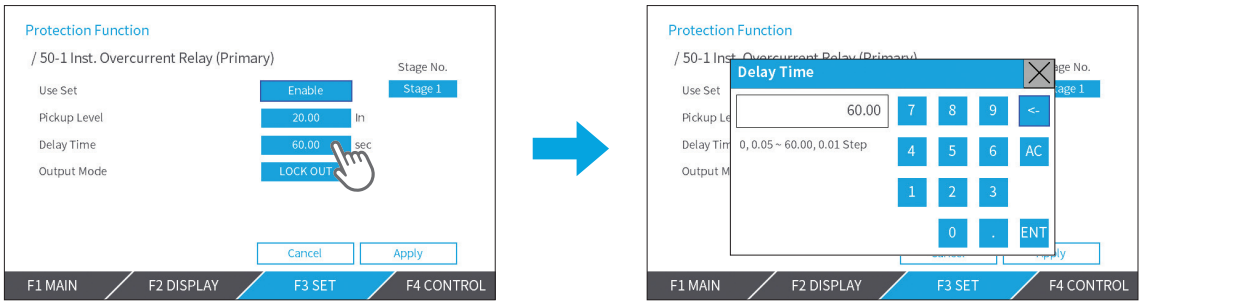
Activation of Protection Relay

When the Set Use option for a protection is set to Enable, the item is highlighted in blue, and the corresponding setting values become editable.



Protection Relay Setting Configuration

To specify a desired setting item, touch the parameter field, then select a value from the drop-down menu according to the parameter type, or use the numeric keypad to enter detailed values.



Technical Data

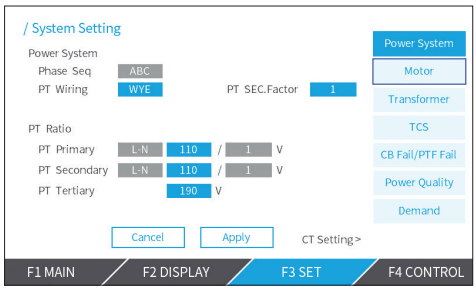
User Interface

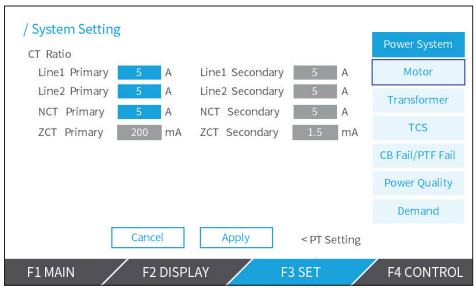
Basic Operation Guide

4. System Setting

The System Setting menu consists of configuration options for the operating environment, including Power System, Motor and Transformer, TCS settings, CB/PTF Fail settings, power quality monitoring, and Demand settings.

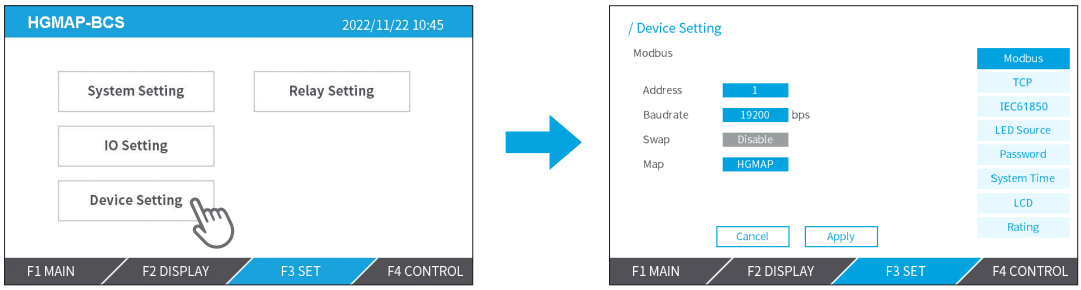
POWER SYSTEM Set





No.	Item		Description	Set range
1	Phase Seq		Phase sequence	ABC, ACB
2	PT Wiring		PT wiring type	WYE, DEL, ADV
3	PT Factor		PT Factor	1, $\sqrt{3}$
4	PT Ratio	PT PRI	PT primary rating (L-N, L-L)	100 – 345,000V (10step)
		PT PRI Factor	PT primary factor	1, $\sqrt{3}$
		PT SEC	PT secondary rating (L-N, L-L)	100 – 190V (10step)
		PT SEC Factor	PT secondary factor	1, $\sqrt{3}$
		PT TER	PT tertiary rating	100 – 190V (10step)
5	CT Ratio	Line1 PRI	Line1 CT primary rating	1 – 9,000A or 5 – 9,000A (5step)
		Line1 SEC	Line1 CT secondary rating	1A, 5A (fixed)
		Line2 PRI	Line2 CT primary rating	1 – 9,000A or 5 – 9,000A (5step)
		Line2 SEC	Line2 CT secondary rating	1A, 5A (fixed)
		NCT PRI	NCT primary rating	5 – 9,000A (1step)
		NCT SEC	NCT secondary rating	1A, 5A
		ZCT PRI	ZCT primary rating	200mA (fixed)
		ZCT SEC	ZCT secondary rating	1.5mA (fixed)

COMMUNICATION Set

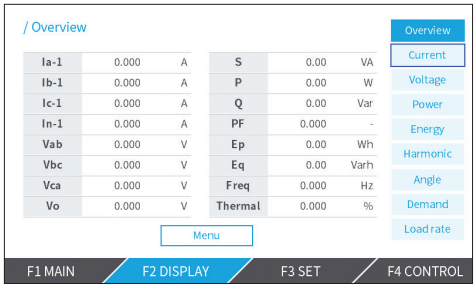


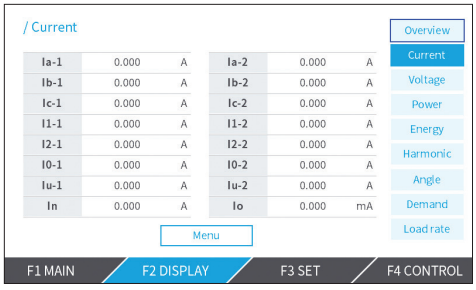
No.	Item	Description	Set range
1	Modbus/RTU – Address	Address set	1 – 247 (1step)
2	Modbus/RTU – Baud rate	Baud rate set (unit: bps)	9600, 19200, 38400, 57600
3	Modbus/RTU – Swap	SWAP function use set	Disable, Enable
4	Modbus/RTU – Map	Address map type set	HGMAP, HiMAP

※ SWAP Function provides a word-swap feature for float variables (measurement data).
For example, when the value 0x1234 5678 is received, enabling SWAP ON will return the data as 0x5678 1234.

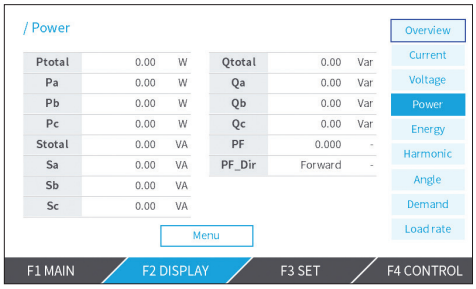
5. Measurement

In the DISPLAY menu, measurement values such as voltage, current, power and energy, harmonics, angle, power factor, and frequency can be monitored.

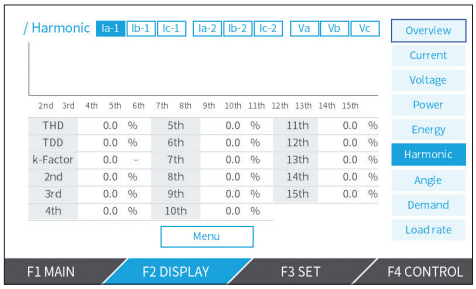


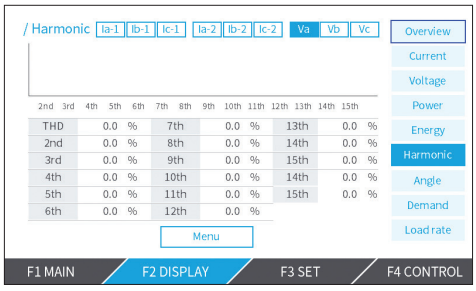




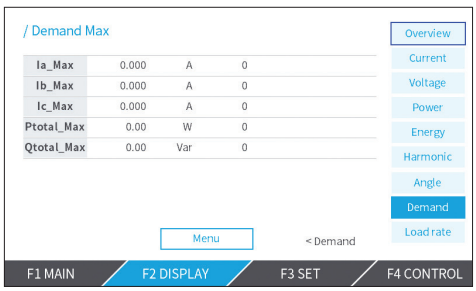


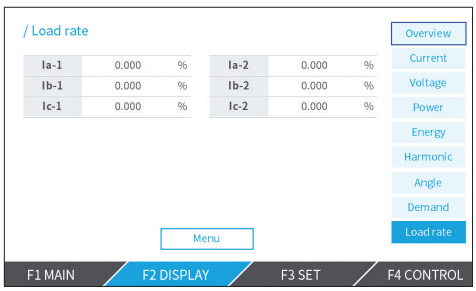












Technical Data

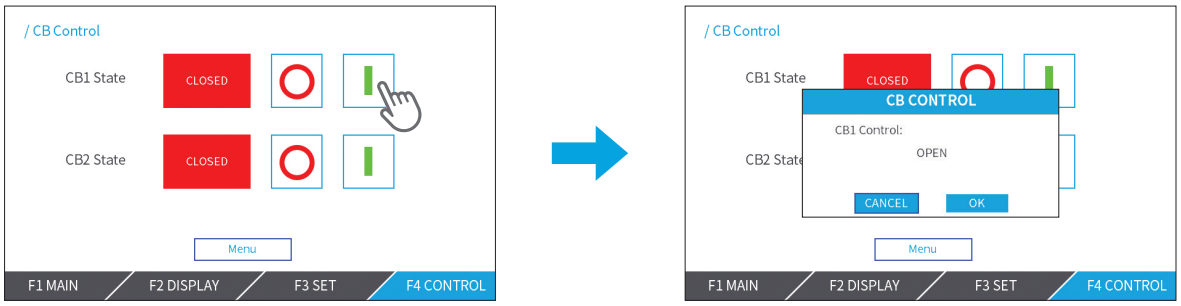
User Interface

HMI Basic Operation Guide

6. Control

Circuit Breaker Control

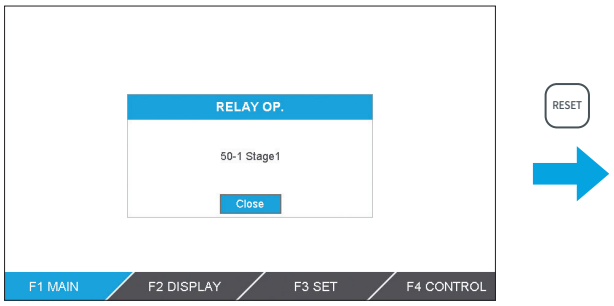
Circuit breakers(CB1, CB2) can be controlled either from the CONTROL menu → CB Control page or through the circuit breaker control buttons on the front.
To operate the breaker from the menu, the control authority must be in the LOCAL mode, and the password must first be unlocked. (When the LOCAL/REMOTE button LED is light in red)



※ To control the breaker remotely, switch to REMOTE mode

FAULT RESET

To reset the DO status(when the DO mode is set to LATCH) and LED after the relay operates due to a system fault, press the RESET button.

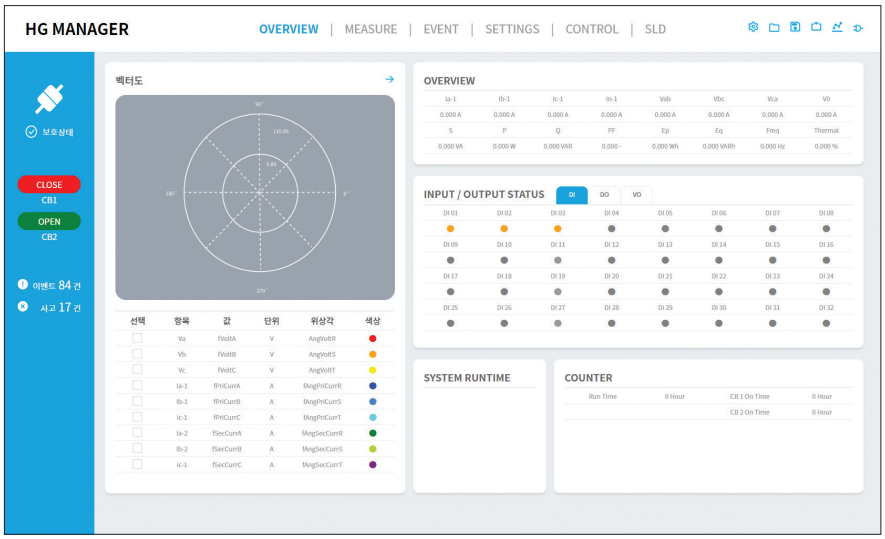


1. Check the trip information.
2. Ensure that the circuit breaker is in OPEN position.
3. Press the “RESET” button to initialize the LCD, LED indicators, and DO outputs.
4. Check the event and fault records on the DISPLAY menu – EVENT/FAULT page.

Notes
- If the root cause of the trip is not resolved and the protection relay is disabled, the fault condition may persist and result in severe damage to the power system.
- RESET will not operate if any activated protection relay continues to satisfy its trip condition.
- If necessary, download the event and fault wave data using the HG Manager software and submit them to Customer Support for detailed root-cause analysis.

HG MANAGER : HGMAP Management Software for PC

HG Manager



Software Features

- HG Manager is a PC-based management software developed exclusively for the HGMAP protection relay.
- Using HG Manager allows users to configure relay settings more conveniently and efficiently after installation.
- It also provides effective monitoring of system conditions, event management, and analysis of fault history and fault waveforms during device operation.
- Relay setting data and various fault-history data can be saved to a PC, and setting values stored on the PC can be directly transferred to the relay, making it convenient for configuring multiple relays.
- A built-in Comtrade viewer enables easy viewing and analysis of fault waveforms without the need for additional waveform-analysis software.
- HG Manager can be downloaded free of charge from the company website under the “Customer Support – Download” section.
  (URL: https://www.hd-hyundaielectric.com/elect/ko/customer/download.jsp)
- For communication, connect the relay’s front USB port to the PC using a USB mini-B cable. The cable is not included by default.

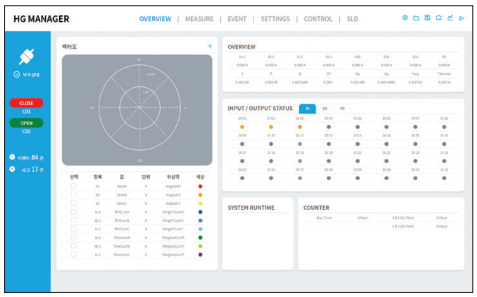
System Requirements

- O/S : Microsoft Windows 7 or later
- Interface : USB 2.0 port (for USB Mini-B cable connection)
- Internet Browser : Compatible with Google Chrome and Microsoft Edge (latest version is recommended)

Technical Data

Features of HG Manager

OVERVIEW



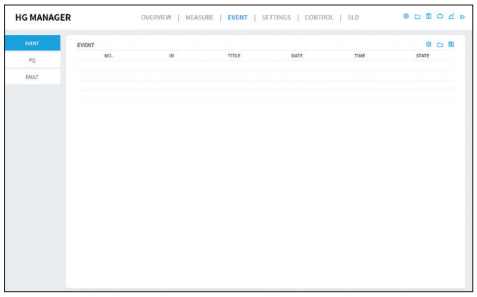
- The relay provides a comprehensive display of the current system and relay status, including real-time phase current and line-to-line voltage measurements, contact status, circuit breaker operation status, and the presence of events or faults.

MEASURE



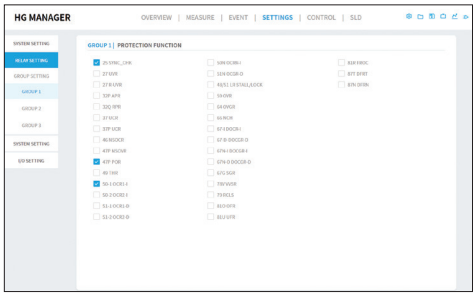
- Detailed electrical data provided by the device such as current, voltage, power, and energy can be monitored along with load conditions and contact operation counts.

EVENT



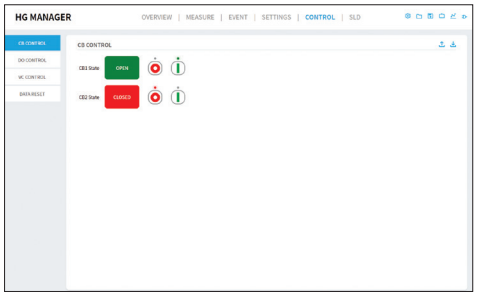
- Internal relay events and fault event records can be reviewed, and detailed fault-current data can be accessed. Fault waveforms can be converted to Comtrade format and viewed using the built-in waveform viewer.

SETTING



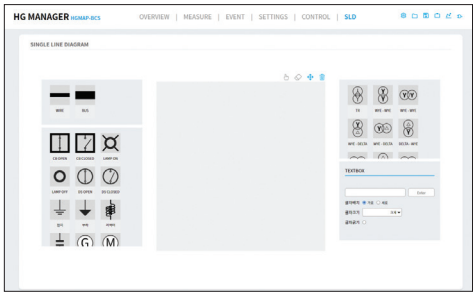
- Various product settings, including system specifications and relay protection relay operating conditions, can be modified. The configuration data can be saved to a PC for use in maintenance or new device installations.

CONTROL



- The operation of the circuit breaker and DI/DO contacts can be controlled, and various internal relay data can be initialized.

SLD



- The single line diagram displayed on the relay's main screen can be freely edited.

※ Information displayed in HG Manager may vary depending on the relay model and specifications

HG CB LINK : HGMAP Management Application for Mobile

HG CB LINK





Software Features

- HG CB LINK is an integrated mobile management application for the HG relay and measurement product lineup.
- Without requiring a separate PC or communication equipment, users can conveniently monitor and configure product status by connecting with a smartphone or tablet PC.
- The application provides an intuitive user interface with screen layouts optimized for various display sizes to ensure easy operation.
- In addition to general status screens, it offers vector diagram, harmonic charts and fault waveform viewer to enable highly effective maintenance and fault cause analysis.
- SLD(single line diagram) editor is included.
- Bluetooth connectivity is required for software operation (supports BLE 4.0 or higher).
- The HG CB LINK application can be downloaded free of charge from Google Play.

System Requirements

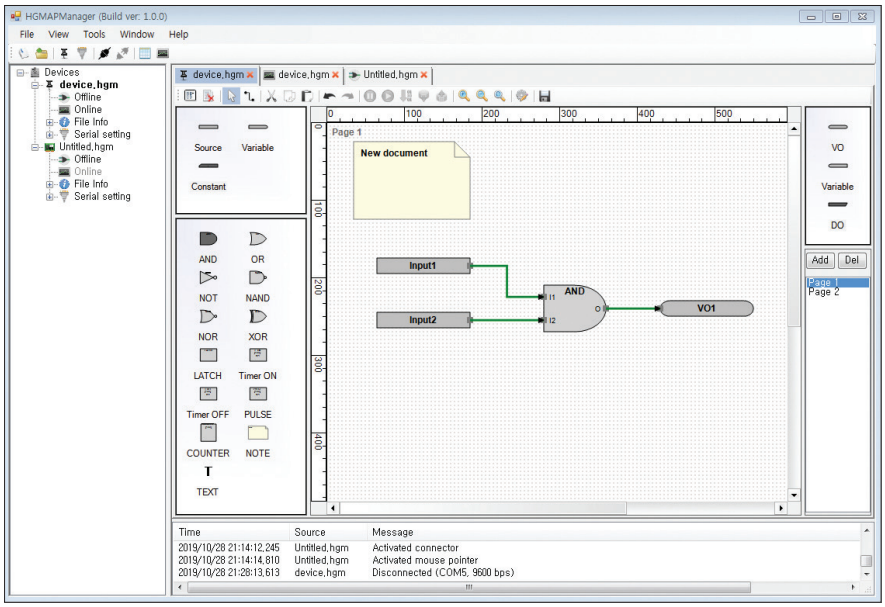
- Recommended : Android Nougat or later on smartphones or tablet PCs.

※ Above screens shown are under development and may be subject to change without prior notice.

Technical Data

PLC Logic Editor

HGMAP PLC Logic Editor



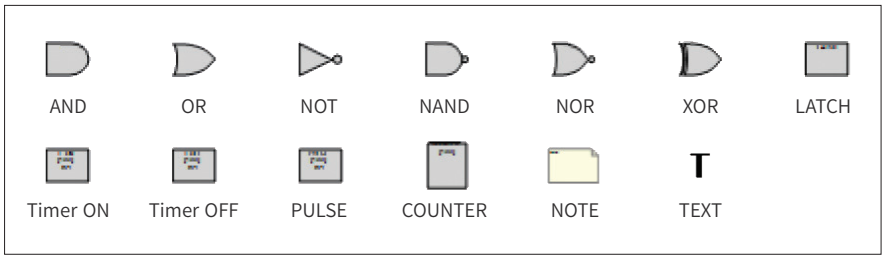
Software Features

- FBD(Function Block Diagram) based PLC Logic Editor
- Provides a graphical symbol interface that allows users to easily understand the logic structure and flow, and facilitates convenient modification of logic.

Key Functions

- Edit tool for user-defined PLC logic
- It supports creating new PLC logic, editing existing logic, and transmitting, receiving, and saving logic data.

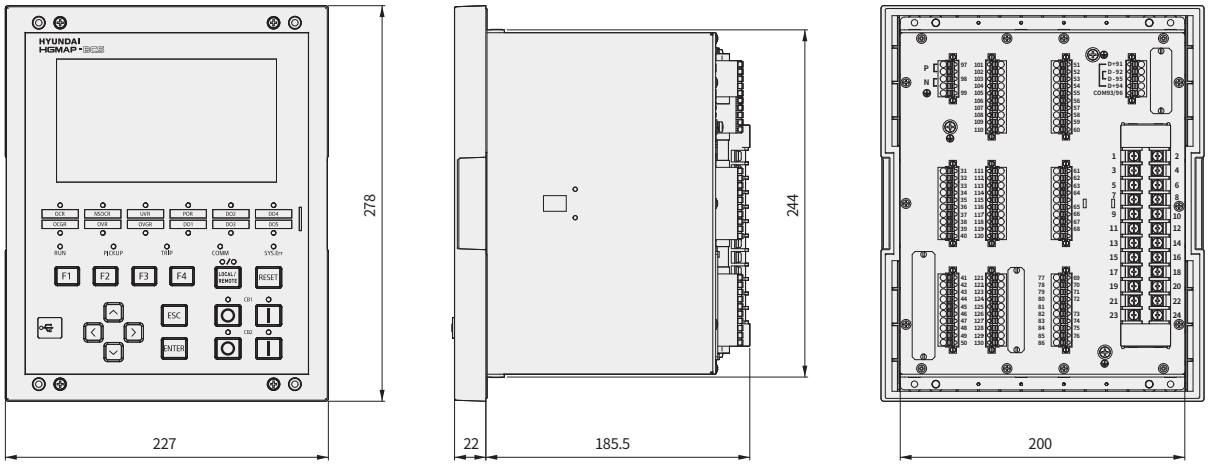
Function Block



Outfit

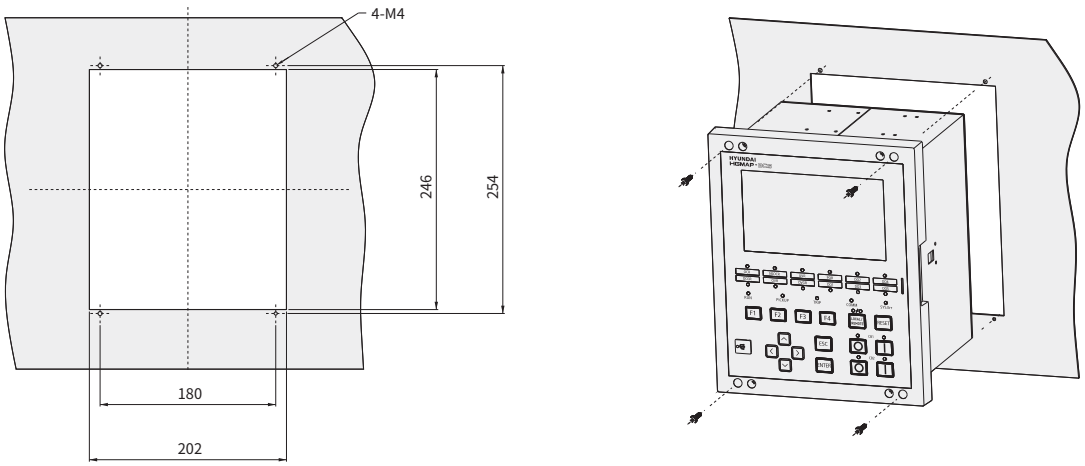
Dimension

unit : mm



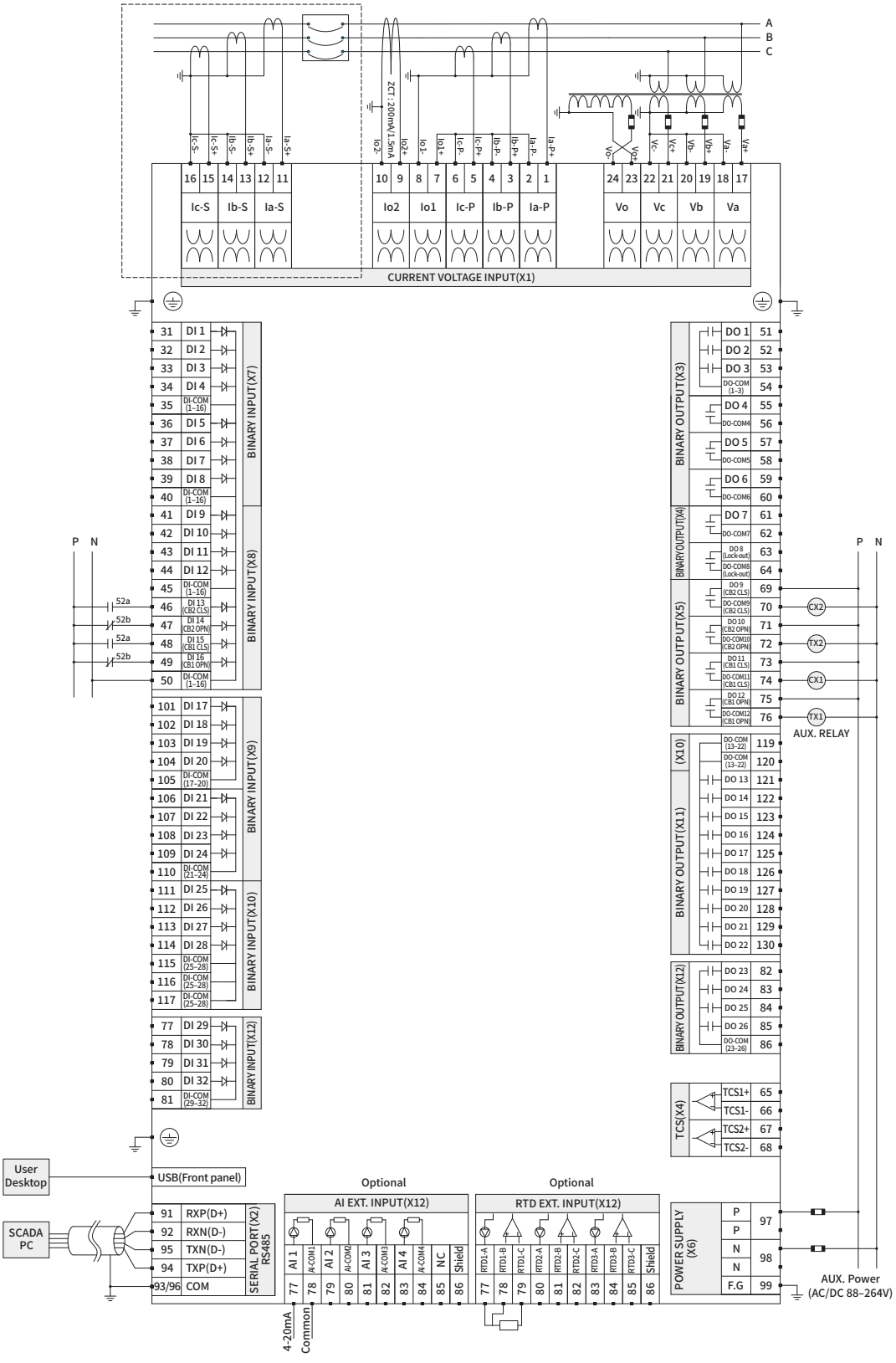
Panel cut

unit : mm



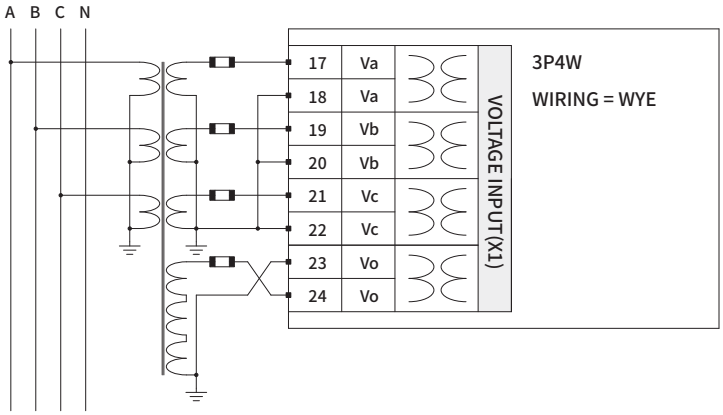
Wiring diagram

3-phase Standard

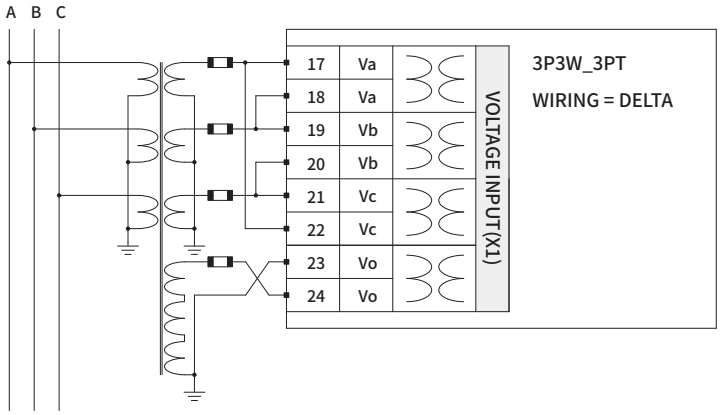


PT Wiring

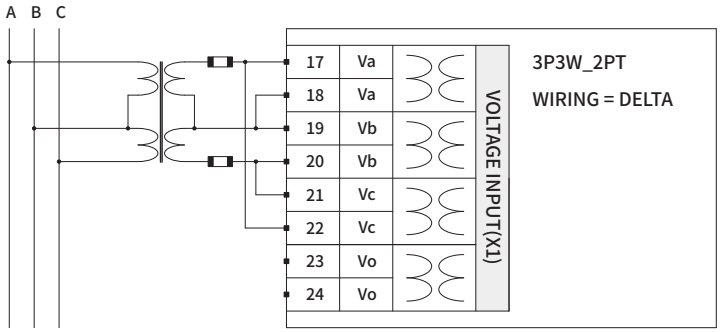
WYE (3P4W)



DELTA (3P3W, 3PT)

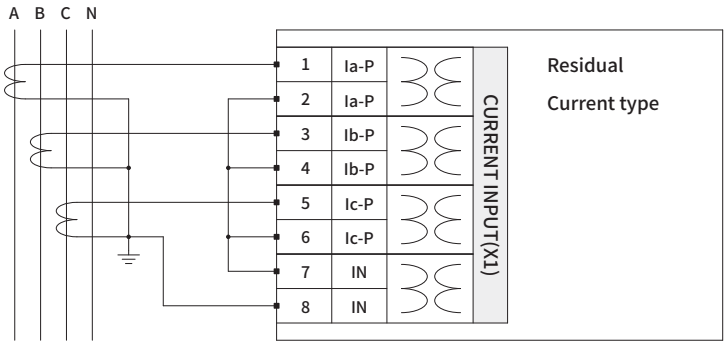


DELTA (3P3W, 2PT)

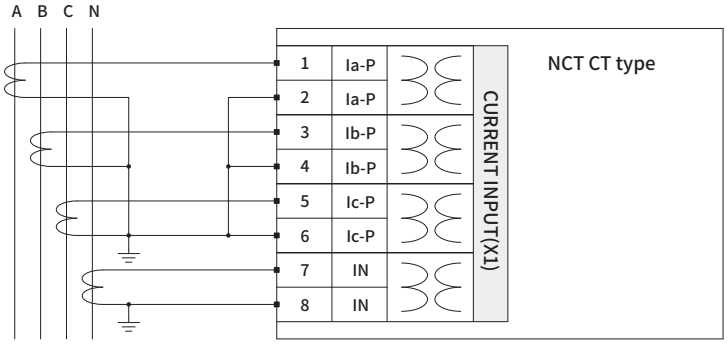


CT Wiring

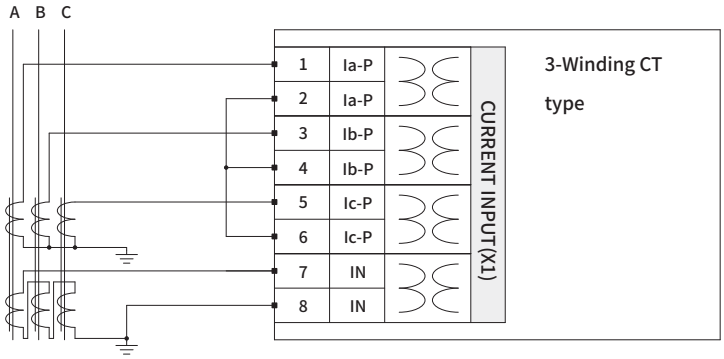
3P4W Residual Current



3P4W NCT

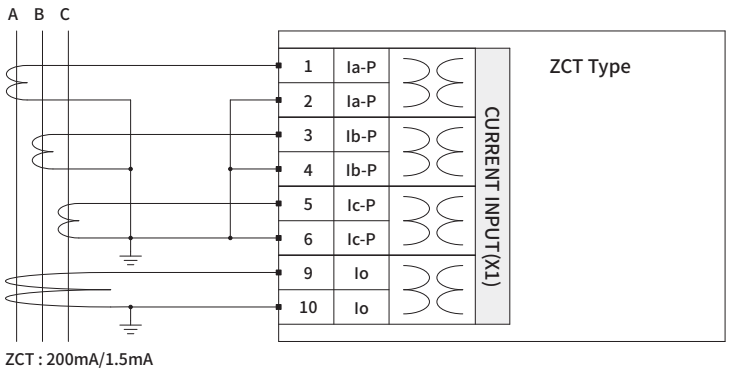


3P3W 3-Winding CT

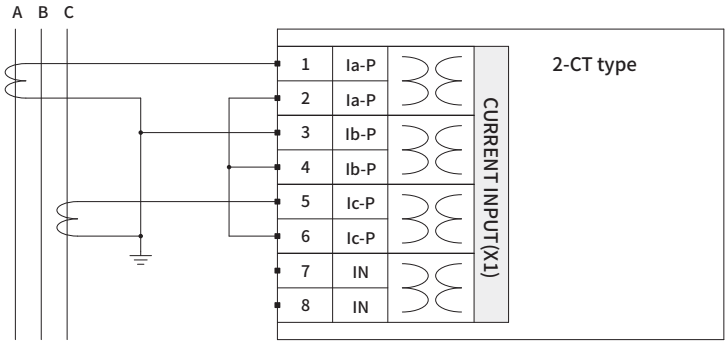


CT Wiring

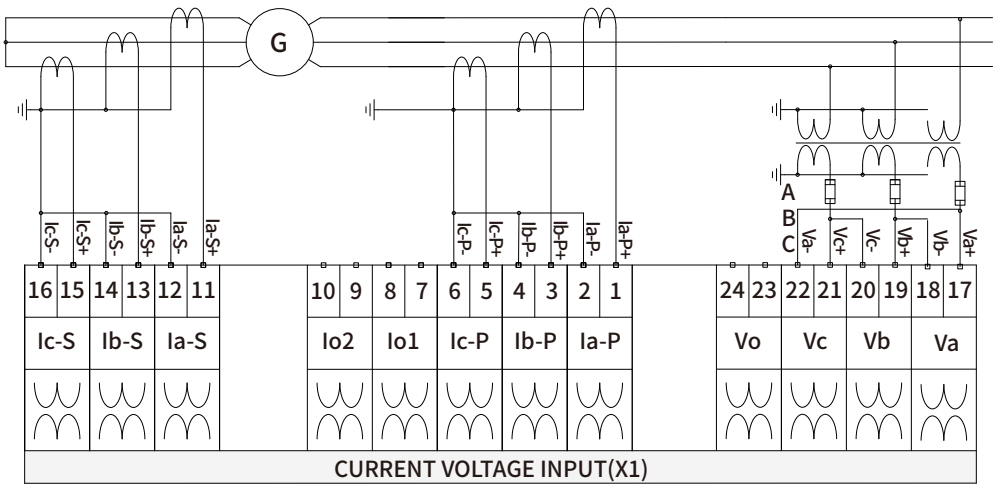
3P3W ZCT



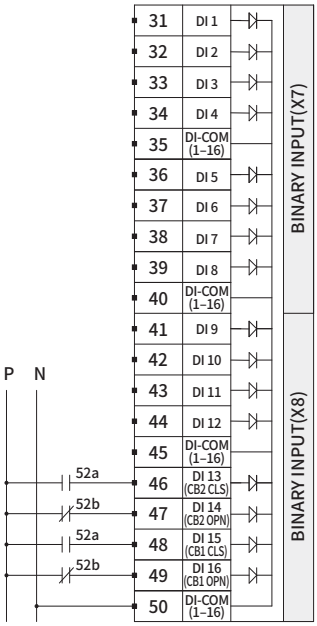
2CT

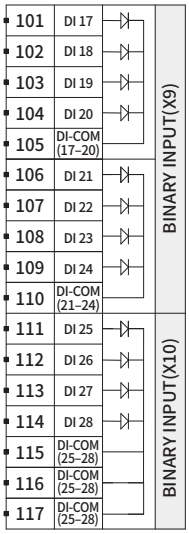


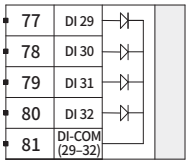
Generator



DI/DO Wiring

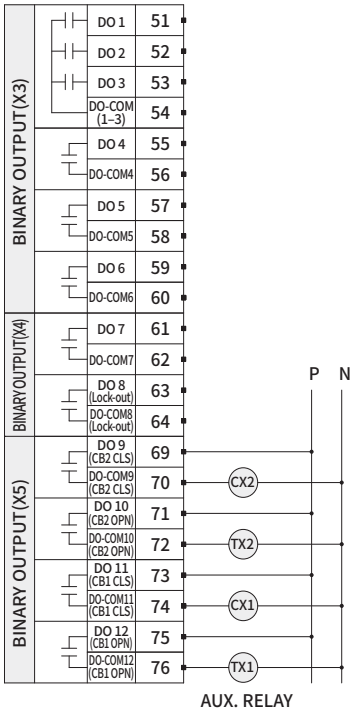


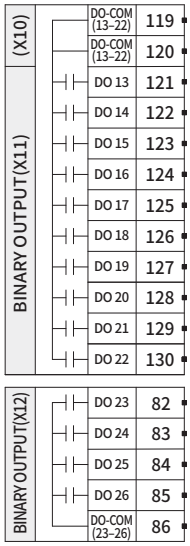




DO source table

No.	Default value
	BCS
DO 01	NONE
DO 02	NONE
DO 03	NONE
DO 04	NONE
DO 05	NONE
DO 06	NONE
DO 07	NONE
DO 08	NONE
DO 09	NONE
DO 10	NONE
DO 11	NONE
DO 12	NONE
DO 13	NONE
DO 14	NONE
DO 15	NONE
DO 16	NONE
DO 17	NONE
DO 18	NONE
DO 19	NONE
DO 20	NONE
DO 21	NONE
DO 22	NONE
DO 23	NONE
DO 24	NONE
DO 25	NONE
DO 26	NONE





Order code

HGMAP-BCS

HGMAP	B	F	N	V	C1	FV
Category	Type	Protection	CT4 Type	Outfit	Communication	Control Power

Category		Type		Protection		CT4 Type		Outfit		Communication		Control Power	
HGMAP	Digital Protection Relay	**B**	Bay Control type	**F**	Feeder/Incoming Motor, TR Renewable(ECO)	**N**	NCT/ ZCT	**V**	Vertical	**C1**	MODBUS/RS485	**FV**	A/DC 88–265V
										C7	TCP/HSR-PRP(Fiber) 1) 2)		
										C9	DNP3.0/RS485		

※ 1) TCP/HSR-PRP(UTP, Fiber) includes MODBUS/RS485 by default.
2) Fiber optic type : Multimode SFP LC type(100Mb)
3) Retrofit option is exclusive for the retrofit of HiMAP-BCS discontinued type.
4) DE(DIO Extension) option : DI 16 channel, DO 14 channel
5) AE(AI Extension) option : DI 12 channel, DO 10 channel, AI(4–20mA) 4 channel
6) RE(RTD Extension) option : DI 12 channel, DO 10 channel, RTD(PT100) 3 channel

F6	A1	A5	NO	NO	NO
Frequency	PT rating	CT rating	Retrofit	Extension	User Option
F6 60Hz	**A1** AC 110V	**A5** AC 5A	**NO** No Option	**NO** No Option	**NO** No Option
F5 50Hz		**A1** AC 1A	**RM** Retrofit [3]	**DE** DIO Extension [4]	**MA** Maritime
				AE AI Extension [5]	
				RE RTD Extension [6]	

Precautions

Handling Precautions

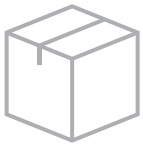
Storage

| Ambient Temperature | -5 to 40°C (average not exceeding 35°C per day)
| Altitude | Below 1,000m above sea level
| Relative Humidity | 45% to 85%

Proper storage conditions must be strictly observed, as the insulation performance and durability of the product may be affected by the storage environment.



- **Do not store the product in locations with high humidity.**
  - The relative humidity must be maintained within the range of 45% to 85%.
- **Avoid areas with excessive dust.**
  - Use covers or packaging materials to prevent dust accumulation, and do not store the product in areas exposed to open air.



- **Store the product in its packaged condition.**
  - Always keep the product stored in its original packaging.
  - Use shelves or similar storage equipment and avoid placing the product directly on the floor.
  - Do not carry heavy items manually; doing so may lead to personal injury.



- **Do not expose the product to direct sunlight for long durations.**



- **Avoid storage in excessively high or low temperatures.**
  - Maintain a storage temperature between - 20°C and +60°C.



- **Do not store the product in environments where corrosive gases are present.**
  - Avoid any exposure to sulfur-containing gases or ammonia gas.

Transportation

⚠ Caution

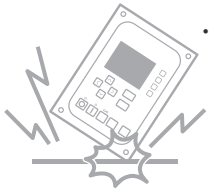
- Do not subject the product to impact during transportation. Dropping the product or applying excessive shock may result in product failure.
- Do not handle the product by its accessories or external lead wires. Doing so may cause injury to the operator or lead to product malfunction.



- **Hold the main body of the product when carrying it.**
  Do not handle the product by the connection cables between the main unit and the display module, nor by any external lead wires.



- **Exercise caution when handling metallic accessories.**
  Sharp surfaces or edges may cause injury.



- **Do not subject the product to impact during transportation.**
  Dropping the product or applying excessive shock may result in product defects or malfunction.



- **Check the product packaging before transportation.**
  Improper or insufficient packaging may cause damage to the product during transit.

Safety Guideline

⚠ Warning

- Ensure that all power source and buses are isolated before installation.
- Do not perform any wiring work when the busbars are energized.
- Never attempt to work by oneself.
- Check all ground path condition to the device before starting.
- Do not handle the device with wet hands.
- All cables must be provided with suitable terminals.
- PT secondary circuit must not be short-circuited.
- CT secondary circuit must not be disconnected.

⚠ Caution

- Any installation and maintenance must be performed by qualified personnel only.
- Check the control voltage condition before wiring.
- Check the rated power and polarity before installation.
- Foreign substances like oils or bolt, metals inside the device can damage the system.
- Be sure to use the auxiliary relay for circuit breaker control.

Storage

- Keep the device in dry and dust-free place away from direct sunlight and moisture.
- Make sure that nothing is broken or covered with soot or pollutant.
- Ensure that there is no damage to terminal block in the rear.
- Verify that the ratings of the PT/CT/ZCT/GPT match the required specifications.

Switchgear

- Be sure to use the auxiliary relay and lockout relay to control the circuit breaker.
- When using magnetic contacts or inductive loads in the control power circuit, be sure to install a clamping diode.
- The device FG ground must be connected to the ground conductor which is electrically isolated from the panel enclosure.

Installation

- Confirm that the voltage of control power meet the specification.
- Ensure that the polarity is correctly connected to the terminal.
- Check that PT, CT secondary is connected well to PT, CT terminal.
- Check that the communication cable is connected to terminal.
- STP(Shielded Twisted Pair) cable is only permitted for a communication.

Operation

- Always check the negative sequence voltage and current, power factor, active power during operation.
- The circuit breaker shall be opened before the change of setting value.
- Unauthorized operation is forbidden.

Test

- Voltage withstand test and insulation resistance test for protection relay are strictly prohibited. If required, keep separate the inner frame of relay from the panel before these tests.
- Before the test, circuit shall be separated from CT, PT secondary using CT test terminal and PT test terminal.
- All test equipment shall undergo proper calibration.
- Check the permissible range and rating for test voltage and current.
- Make sure that registered setting value is accordance with the setting table.
- During the protection test, all function which is not subject to the test shall be disabled.

Maintenance and Replacement

- Follow the detailed work order.
- Do not replace the device in rainy, humid condition.
- Use the PT test terminal and CT test terminal.
- Ensure that the relay is clearly de-energized before maintenance.
- Keep a setting value in writing before replacement.
- Clear and conspicuous consent from site manager is required for any site work.
- Check that control power is clearly off before replacement.

Certificate

HGMAP-BCS



면제번호 : 제 2022-바-023 호

Korea Electrical Manufacturers Association

공 인 검 수 시 험 면 제 증

회 사 명 : 에이치디현대일렉트릭㈜

대 표 자 : 조석

소 재 지 : 경기도 안성시 서운면 제3동단3길 53

표 시 품 목

품 목 명 : 보호계전기

제 품 명 : 복합형계전기 (디지털형)

정 격 : AC(110, 190)V AC 5A AC 1.5mA 60Hz 제어전원: AC 및 DC(88 ~ 265)V
(HGMAP-BCS)
※ 계전요소: DFR-87T, DFR-87N, 50, 51, 50N, 51N, 59, 64, 27, 27R, 37, 67I, 67D, 67NI, 67ND, 67G, 46, 47N, 47P, 32P, 32Q, 37P, 81O, 81U, 81R, 49, 48/51R, 25, 79, 66, 78V

적용표준 : SPS-KEMC 1120-0579 (2018)

전기기기 공인시험기준 및 방법에 관한 요령(산업통상자원부 고시 제2021-34호) 제7조의 규정에 의하여 위와 같이 공인검수시험 면제대상임을 확인합니다.

발 행 일 : 2023년 4월 10일 월요일

면제 기한 : 2025년 10월 14일 화요일

한국전기산업진흥회

HGMAP-A



면제번호 : 제 2021-바-034 호

Korea Electrical Manufacturers Association

공 인 검 수 시 험 면 제 증

회 사 명 : 에이치디현대일렉트릭㈜

대 표 자 : 조석

소 재 지 : 경기도 안성시 서운면 제3동단3길 53

표 시 품 목

품 목 명 : 보호계전기

제 품 명 : 복합형계전기 (디지털형)

정 격 : AC 5A, AC(110,190)V, Aux.power: AC(88~265V) or DC(88~265V) (HGMAP-A-N)
계전요소 : 50, 51, 50N, 51N, 59, 64, 27, 67I, 67D, 67NI, 67ND, 37, 46, 47N, 47P, 32P, 32Q, 37P, 81O, 81U, 81R, 49, 48/51R, 66, 25, 78V

적용표준 : SPS-KEMC 1120-0579 (2018)

전기기기 공인시험기준 및 방법에 관한 요령(산업통상자원부 고시 제2021-34호) 제7조의 규정에 의하여 위와 같이 공인검수시험 면제대상임을 확인합니다.

발 행 일 : 2023년 4월 10일 월요일

면제 기한 : 2024년 8월 9일 금요일

한국전기산업진흥회



면제번호 : 제 2021-바-035 호

Korea Electrical Manufacturers Association

공 인 검 수 시 험 면 제 증

회 사 명 : 에이치디현대일렉트릭㈜

대 표 자 : 조석

소 재 지 : 경기도 안성시 서운면 제3동단3길 53

표 시 품 목

품 목 명 : 보호계전기

제 품 명 : 복합형계전기 (디지털형)

정 격 : AC 5A AC 1.5mA AC(110/190)V Aux.power: AC(88~265V) or DC(88~265V) (HGMAP-A-Z)
계전요소 : 50, 51, 59, 64, 27, 67I, 67D, 67G, 37, 46, 47N, 47P, 32P, 32Q, 37P, 81O, 81U, 81R, 49, 48/51R, 66, 25, 78V

적용표준 : SPS-KEMC 1120-0579 (2018)

전기기기 공인시험기준 및 방법에 관한 요령(산업통상자원부 고시 제2021-34호) 제7조의 규정에 의하여 위와 같이 공인검수시험 면제대상임을 확인합니다.

발 행 일 : 2023년 4월 10일 월요일

면제 기한 : 2024년 8월 9일 금요일

한국전기산업진흥회

HGMAP-S



면제번호 : 제 2020-바-036 호
Korea Electrical Manufacturers Association
공 인 검 수 시 험 면 제 증

회 사 명 : 에이치디현대일렉트릭㈜
대 표 자 : 조석
소 재 지 : 경기도 안성시 서운면 제3동단3길 53

표 시 품 목
품 목 명 : 보호계전기
제 품 명 : 복합형계전기 (디지털형)
정 격 : AC 5A AC(110, 190)V Aux.power : DC 110V or DC 125V (HGMAP SF N)
계전요소 : 50, 51, 50N, 51N, 59, 64, 27, 67ND, 46, 47N, 47P, 49, 48/51R, 37, 66
적용표준 : SPS-KEMC 1120-0579 (2018)

전기기기 공인시험기준 및 방법에 관한 요령(산업통상자원부 고시 제2021-34호) 제7조의 규정에 의하여 위와 같이 공인검수시험 면제대상임을 확인합니다.

발 행 일 : 2023년 4월 10일 월요일
면제기한 : 2023년 12월 10일 일요일

한국전기산업진흥회



면제번호 : 제 2020-바-038 호
Korea Electrical Manufacturers Association
공 인 검 수 시 험 면 제 증

회 사 명 : 에이치디현대일렉트릭㈜
대 표 자 : 조석
소 재 지 : 경기도 안성시 서운면 제3동단3길 53

표 시 품 목
품 목 명 : 보호계전기
제 품 명 : 복합형계전기 (디지털형)
정 격 : AC 5A AC(110, 190)V Aux.power : DC 110V or DC 125V (HGMAP SF RN)
계전요소 : 50, 51, 50N, 51N, 59, 64, 27, 67ND, 37, 46, 47N, 47P, 49, 48/51R, 66, 25
적용표준 : SPS-KEMC 1120-0579 (2018)

전기기기 공인시험기준 및 방법에 관한 요령(산업통상자원부 고시 제2021-34호) 제7조의 규정에 의하여 위와 같이 공인검수시험 면제대상임을 확인합니다.

발 행 일 : 2023년 4월 10일 월요일
면제기한 : 2023년 12월 10일 일요일

한국전기산업진흥회



면제번호 : 제 2020-바-002 호
Korea Electrical Manufacturers Association
공 인 검 수 시 험 면 제 증

회 사 명 : 에이치디현대일렉트릭㈜
대 표 자 : 조석
소 재 지 : 경기도 안성시 서운면 제3동단3길 53

표 시 품 목
품 목 명 : 보호계전기
제 품 명 : 복합형계전기 (디지털형)
정 격 : AC 5A Aux. Power : DC 110V or DC 125V (HGMAP ST N)
보호요소 : OCR(50), OCR(51), OCGR(50N), OCGR(51N), DFR(87T, 87N1, 87N2)
적용표준 : SPS-KEMC 1120-0579 (2018)

전기기기 공인시험기준 및 방법에 관한 요령(산업통상자원부 고시 제2021-34호) 제7조의 규정에 의하여 위와 같이 공인검수시험 면제대상임을 확인합니다.

발 행 일 : 2023년 4월 10일 월요일
면제기한 : 2026년 1월 10일 토요일

한국전기산업진흥회



면제번호 : 제 2019-바-015 호
Korea Electrical Manufacturers Association
공 인 검 수 시 험 면 제 증

회 사 명 : 에이치디현대일렉트릭㈜
대 표 자 : 조석
소 재 지 : 경기도 안성시 서운면 제3동단3길 53

표 시 품 목
품 목 명 : 보호계전기
제 품 명 : 복합형계전기(디지털형)
정 격 : AC 110V 5A 제어전원 : DC 110V 및 125V (HGMAP-SE-N)
보호요소 : OCR, OCGR, OVR, UVR, OVGR, DOCR, DOCGR, APR, RPR, UPR, OFR, UFR, FROC
적용표준 : SPS-KEMC 1120-0579 (2018)

전기기기 공인시험기준 및 방법에 관한 요령(산업통상자원부 고시 제2021-34호) 제7조의 규정에 의하여 위와 같이 공인검수시험 면제대상임을 확인합니다.

발 행 일 : 2023년 4월 10일 월요일
면제기한 : 2025년 5월 31일 토요일

한국전기산업진흥회

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HD HYUNDAI ELECTRIC

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